AEROTHON 2026

SAE INDIA AEROSPACE

DESIGN REPORT

TEAM VAYURATHA

DHRUV TARA OF SKY

REG. NO. AT2025069



DESIGN • INNOVATE • ELEVATE

DYP DPU

Dr. D.Y. Patil Institute of Technology
Pimpri, Pune - 411018

AEROTHON 2026

SAEINDIA AEROSPACE FORUM

APPENDIX A

STATEMENT OF COMPLIANCE

Certification of Qualification

Team Name: TEAM VAYURATHA

University/Institute: DR D. Y. PATIL INSTITUTE OF TECHNOLOGY, PIMPRI, PUNE

Faculty Advisor: DR. RAVI. SINHA

Faculty Advisor's Email: ravi.sinha@dypvp.edu.in

Statement of Compliance:

As Faculty Advisor, I certify that the registered team members are enrolled in collegiate courses. This team has designed the UAS for the SAE AEROTHON 2026 contest, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.

Dr. Ravi Sinha

Signature of Faculty Advisor

24/05/26

Date

Team Captain Information:

Team Captain's Name: TANUJ GAVRKAR

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Abstract

This report presents the design process and key decisions made by our team, **"Vayuratha"**, during the development of our unmanned aerial system for **SAE AEROTHON 2026**. We see this work as a humble contribution towards advancing autonomous systems for public safety, national security, and surveillance.

The name **"Vayuratha"** is inspired by the resilience of a chariot in motion, reflecting our philosophy of reliability in the air. Our drone, **"Dhruv"**, is named after the pole star, a constant guide for travellers, wanderers & seekers long before modern navigation. It stands as a tribute to the mentors who have guided, challenged, and shaped our journey.

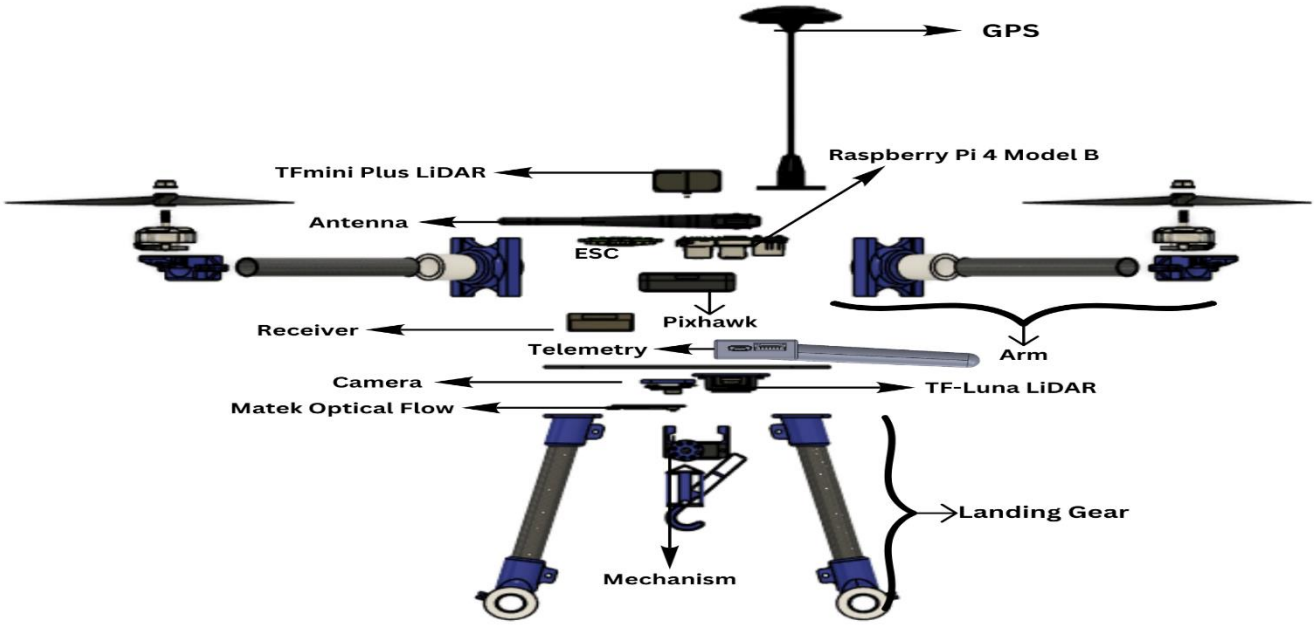


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1.0 Technical Content

1.1 Conceptual Design

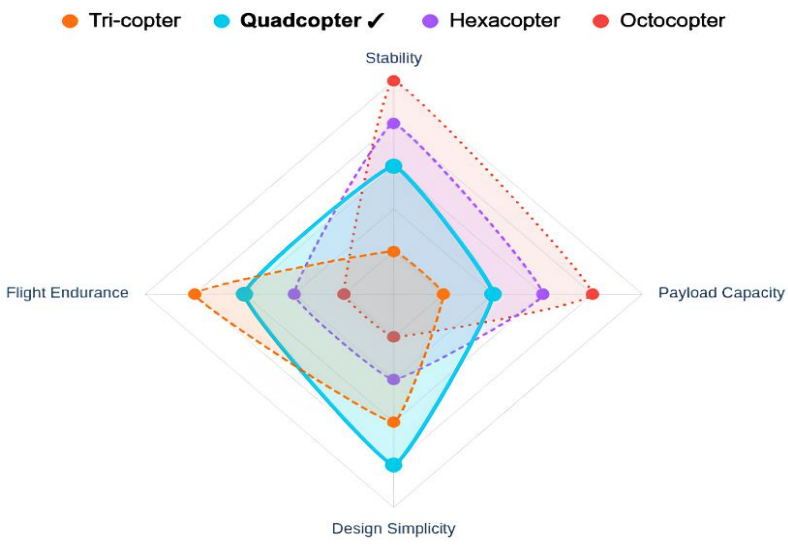


To execute the required problems, The Design philosophy for our Drone is **"Refined, Reliable, Resilient"**

Technically speaking, a drone such as this must have:

1) Reliable configuration built on strong core design fundamentals.
2) Protected airframe with a proper body cover/canopy.
3) Robust, well-tested electronics backed by datasheets and simulations.
4) Stability at core: prop selection, CG, and everything calculated and justified.
5) Built-in redundancy with a high factor of safety (>4) and fail-safe

1.1.1 Rotorcraft Selection



These **four metrics** were selected based on the actual demands of the competition missions.

1) **Stability** is compulsory for accurate hover and navigation, and the quadcopter's symmetric layout provides reliable control without added complexity.

2) **Payload** demand is low (100 g within 2 kg MTOW), making higher rotor configurations unnecessary.

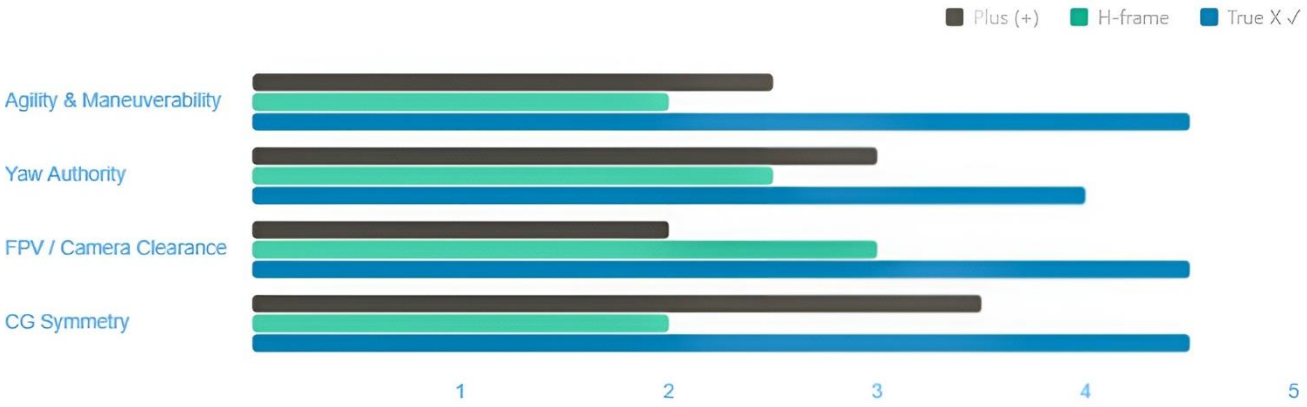
3) **Design simplicity**: fewer motors lead to easier tuning & better Yaw management.

4) For **endurance**, a quadcopter offers a balanced power profile, unlike power-hungry 6 or 8 rotors.

Overall, the **quadcopter** achieves the right balance, meeting mission requirements without unnecessary complexity or overdesign.

1.1.2 Configuration Selection

A True X configuration was chosen because of the following analysis.



1) **Agility & manoeuvrability:** True X gives symmetric roll/pitch control for tight obstacle navigation and is best for pushing through a corridor.
2) **Yaw authority:** all four motors contribute evenly in True X, giving cleaner and more precise yaw control
3) **FPV / camera clearance**: unobstructed forward view in True X, necessary for observation and QR-based tasks
4) **CG symmetry**: symmetric arm layout ensures a stable Centre of Gravity.

1.2 Product Benchmarking

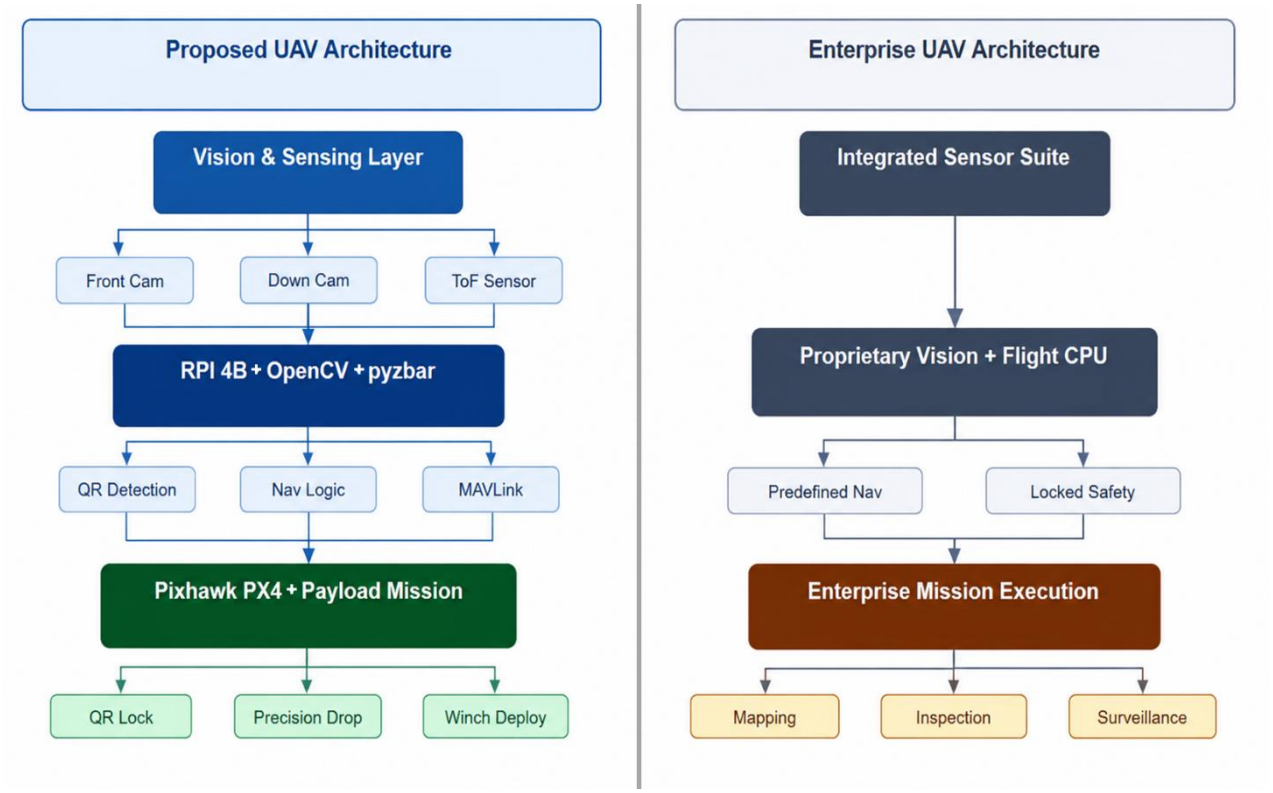
1.2.1 Comparative Analysis: Deployment and Overall Architecture

The benchmarking study is designed to compare the proposed UAV with commercial enterprise UAV platforms in relation to autonomous flexibility, payload adaptability, modularity, and operational reliability. The comparison is focusing on system configurability, autonomous operation capability, ease of maintenance, and payload integration rather than compact foldable architecture or consumer portability.

Parameter	Dhruv UAV Design	DJI Mavic 3 Enterprise	Holybro X500 V2
System Architecture	Modular open-frame design	Compact proprietary foldable design	Modular open-frame design
Flight Controller	Pixhawk PX4-based controller	Proprietary DJI controller	Pixhawk PX4-based controller
Onboard Processing	Raspberry Pi 5	Proprietary enterprise processor	Companion computer support

Autonomous Capability	YOLO, OpenCV, QR detection, payload targeting	Enterprise waypoint autonomy	ROS2/PX4 autonomous development
Payload Integration	Custom payload mechanism support	Limited certified payload ecosystem	Modular payload mounting
System Customization	High open-source flexibility	Limited SDK-level access	High open-source flexibility
Maintenance Accessibility	Easy access to electronics and wiring	Compact integrated structure	Easy modular maintenance
Mission Focus	Autonomous payload delivery and obstacle navigation	Enterprise inspection and mapping	Autonomous research and development

1.2.2 Comparative Analysis: Autonomous System Architecture



In contrast to enterprise UAV systems such as DJI Mavic 3 Enterprise that operate on closed proprietary architectures, the proposed UAV is based on an open source modular Pixhawk PX4 and Raspberry Pi system. It provides a simple way to directly integrate custom OpenCV and pyzbar vision pipelines, QR based navigation, MAVLink communication and mission specific payload deployment logic. The architecture is built for autonomous flexibility, simpler subsystem-level debugging, and strong mission adaptability for AeroTHON operations.

1.3 Preliminary Weight Estimation & Thrust Analysis

This section gives the weight estimation of components used in the drone and gives the total calculated weight of each component. It also helps understand the weight distribution, capacity stability and structural requirement.

Sr. No.	Component	Weight Per Unit (g)	No. of Units	Total Weight (g)
1	Frame	350	1	350
2	Battery	500	1	500
3	Motor (Emax ECO II)	50	4	200
4	Companion Computer (Raspberry Pi 4 Model B)	60	1	60
5	Flight Controller (Pixhawk 2.4.8)	40	1	40
6	Front Facing Camera	50	1	50
7	Downward Facing Camera	15	1	15
8	Power Module	25	1	25
9	GPS Module	35	1	35
10	4 in 1 ESC	15	1	15
11	Receiver	15	1	15
12	Telemetry radio	24	1	24
13	Wiring Cables	200	1	200
14	Payload	100	1	100
15	Antenna	20	1	20
16	Screws	1	60	60
17	Component Mount	50	1	50
18	Mechanism	20	1	30
19	Landing gear	100	1	100
20	Propeller	10	4	40
21	Sensors	20	1	20
	Total		72	**1939**

Hence, the estimated preliminary weight of our UAV is: 1939g

The 2:1 equivalent thrust-to-weight ratio has emerged as a common baseline because it generally provides enough reserve thrust for most common drone operations, including surveillance, videography, and corridor navigation.

Required Equivalent Thrust = 2 x Total Weight = 2 x 1939g = **3878g**

For a Quadrotor, Thrust required per Motor = 3850 / 4 = **969.5g**
Thus, a propulsion system must be selected that can provide **3878g** equivalent thrust for stable & efficient manoeuvring.

1.4 Propulsion System Selection

The propulsion system is the basis of the Drone's entire scientific idea of flight; The idea is to propel a mostly non aerodynamic object vertically upwards completely opposing the gravity.

For a micro sized drone an electrical propulsion system is optimal. The propulsion system consists of propellers, Motors & ESCs which are selected in the mentioned order.

1.4.1 Propellers:

It converts the rotary motion into linear thrust to generate lift.



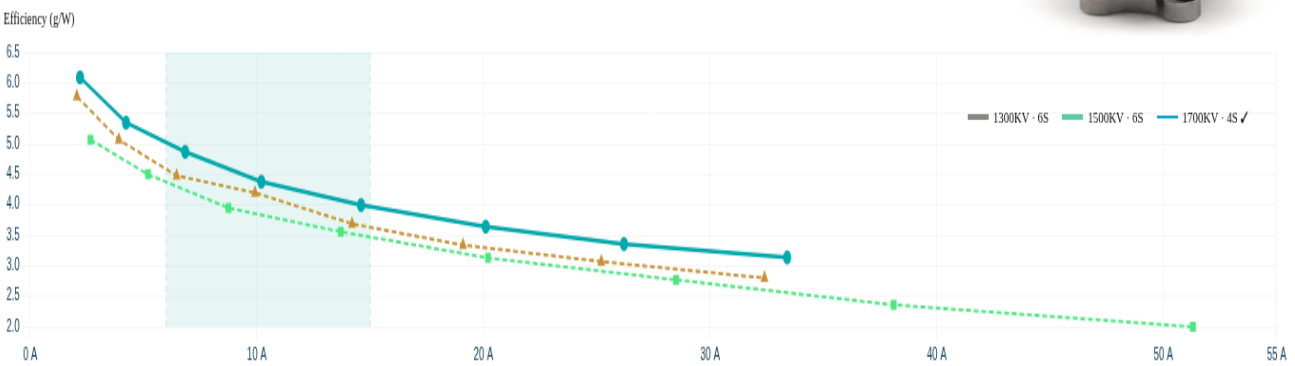
- Our choice of propellers is **a 7x4x3 tri-blade propeller**. 3 blades are chosen because they provide higher thrust at lower throttle, allowing for minute and precise response during corridor navigation. A small size (7”) is chosen because of its utility during missions involving obstacle avoidance which would benefit from a smaller wheelbase.
- A 7” dual blade of similar pitch was considered, and it would yield improved endurance. However, it was a trade-off between better control and flight time, and because of the very specific mission requirements of an obstacle-filled flight path we chose better control.
- Material to be considered is Nylon, which provides strength with light weight.

1.4.2 Motor: ECO II Series Brushless Motor

The Motor is the main actuator driving the entire UAV and must provide the required thrust with the selected choice of propellers. The most important parameter to resolve was the KV rating of the motor. A too high KV rating leads to very high RPM which causes undesirable vibrations. A too low KV rating on the other hand would not provide the necessary throttle response with precision.

The 1300–1700KV range was considered due to its compatibility with 7-inch propellers. The main configurations evaluated were 1300/1500KV with 6S and **1700KV with 4S**. Based on datasheet efficiency analysis, the 1700KV 4S setup showed better efficiency in the 6–10A hover operating range (600–900 g thrust) while also reducing battery weight by nearly 150–250 g compared to a 6S system.





The alternatives which feature 1700KV rating, iFlight XING2 and T-Motor Velox motors feature premium components like titanium shafts and ultra-responsive rotor designs

optimized for FPV racing and aggressive throttle response. However, these features do not provide much of an operational advantage over the EMAX ECO II 2207 1700KV for the proposed application which requires stable hover.

Feature	EMAX ECO II 2207 1700KV	iFlight XING2 2207 1750KV	T-Motor Velox V2207 1750KV	SpeedyBee 2207 1960KV
7” Prop Compatibility	5	5	5	3
Efficiency	5	4	4	3
Torque Output	4	5	5	3
Price	5	3	4	5
Final Score	**19**	**17**	**18**	**14**

1.3.3 Electronic Stability Control: - DYS ARIA 4 in 1 ESC

The DYS ARIA 4-in-1 ESC was selected because of its lightweight integrated design and high-speed digital communication capability.

The ESC weighs only ~15 g, whereas four individual ESCs can weigh nearly 100 - 120 g combined, significantly increasing overall system weight and wiring complexity. The ESC also supports DShot1200 communication, enabling faster throttle signal updates and lower control latency during autonomous navigation and obstacle manoeuvring compared to similar 4 in 1 ESCs such as Racestar Rev35 which support fewer Dshot variants.

Parameter	DYS ARIA 4-in-1 ESC
Continuous Current	60 A
Battery Compatibility	4S LiPo
Firmware	BLHeli_32
Supported Protocols	DShot150/300/600/1200



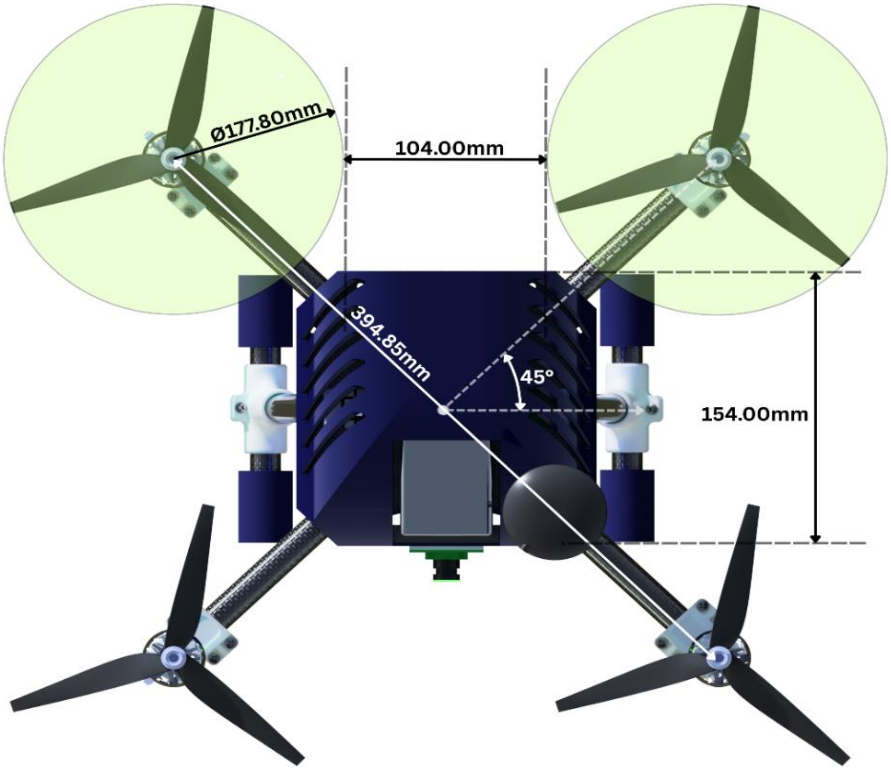
1.5 UAV Sizing:

The sizing of Uncrewed Aircraft Vehicles (UAVs) involves determining the specifications and characteristics of the UAV to meet the requirements of a particular mission or application. This section specifies the main dimensions and UAV metrological specifications.

1.5.1 Wheelbase & Propeller Clearance

Wheelbase is the diagonal centre to centre distance between two diagonal motors. A longer wheelbase generally enhances stability at the potential cost of agility, while a shorter one improves agility but may compromise stability. For the provided mission statement involving close quarter flying we have decided to go with a small wheelbase nearing 400mm instead of the conventional 450-500mm wheelbase generally used for survey applications.

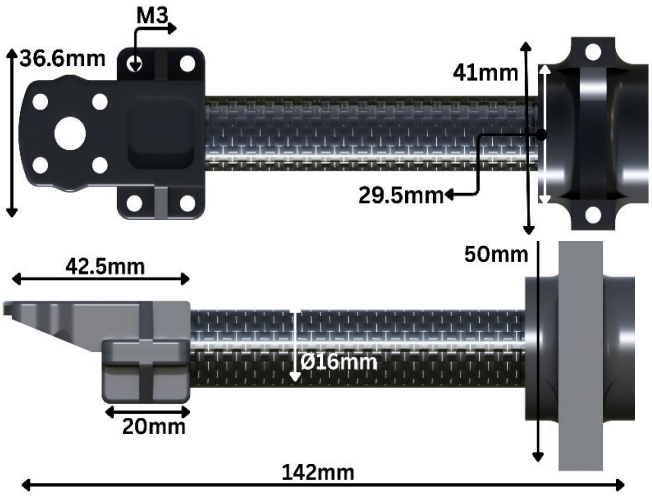
Propeller clearance refers to the essential distance between the propeller blades and any surrounding structures or objects. None of the Drone components except the Motor mounts and Arm rods are kept directly and closely beneath the propellers to minimize propeller wash. The propellers are at a safe distance from any structural component.

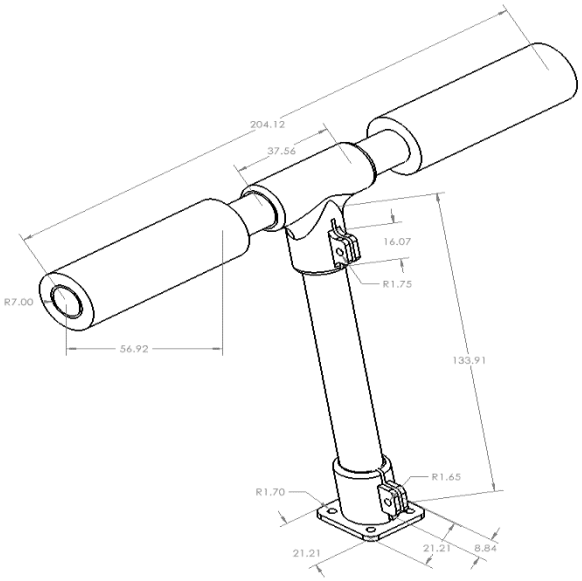


1.5.2 Propeller arms and Landing gear:

The Propeller arms, made of 3K Roll Wrapped Carbon fibre are the main load bearing component of the system. They serve the function of supporting the reaction force that lift the UAV while also maintaining safe clearance of the main frame from the propellers.

The Arms are kept in a pure X configuration as it provides greater stability to counteract the stability issues that may potentially arise due to use of high KV motors and smaller propellers. A TPU roll of 3mm thickness is placed between the arm holder and the rod to dampen vibrations produced at the joints.

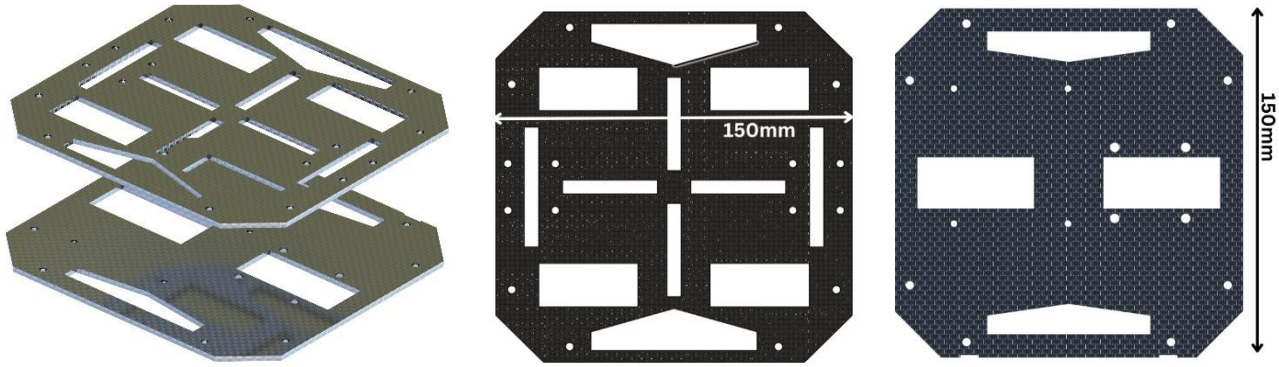




Landing Gear is the impact load bearing component that allow the drone to land without the electronics or the main structure coming into direct contact with the surface. They allow clearance between the payload mechanism and the ground and should sustain the force exerted during landing. The landing gears are tilted at an angle of 12.5 degrees for maximum ground coverage.

1.5.3 Central Plates

The central plates house the avionics and all other subsystems. They are the core structure centre of the drone. They must also sustain the heavy load of the battery and the lift force transmitted through the arms. The lower plate supports the controller and receiver while the upper plate supports the Battery, Companion computer, GPS and the ESC. Both the plates have a square profile with same overall dimensions.



1.6 UAV Performance:

UAV performance analysis involves factors like power required estimation, efficiency, power system (battery) selection, and endurance estimation.

1.6.1 Power Required Estimation:

Name	Voltage (V)	Current (A)	Power (W)
BLDC Motors	14.8	24 A (6 A each)	414.4
4-in-1 ESC	14.8	0.2 A	2.96
Raspberry Pi	5	2 A	15
Raspberry Pi Camera	3.3	0.3 A	1
Global Shutter Camera	5	0.25 A	1.25
Pixhawk FC	5	2 A	10
WiFi Antenna	5	0.5 A	2.5
MG90 Servo Motor	5	0.5 A	2.5
NEO M8N GPS Module	3.3	0.03 A	0.1
Time of Flight Sensor (2)	5	0.08 A (0.04 each)	0.4
Optical Flow Sensor	5	0.15 A	0.75
Total Consumption		**30.01 A**	**450.86 W**

1.6.2 Power System (Battery) Selection:

The selection of the power system, notably the battery, demands meticulous attention to detail. First was the choice of battery chemistry. Out of the available choices Li-Po scored the highest due to its overall well-balanced specifications, its ability to provide burst current required for large corrective manoeuvrers due to its high C rating and its good market availability.

Parameter	Li-Po	Li-Ion	NiMH	Lead Acid
Energy Density	4 / 5	5 / 5	2 / 5	1 / 5
Discharge Capability	5 / 5	3 / 5	2 / 5	1 / 5
Weight Efficiency	4 / 5	5 / 5	2 / 5	1 / 5
Peak Current Delivery	5 / 5	3 / 5	2 / 5	1 / 5

Parameter	Value
Battery Capacity	5200mAh = **5.2Ah**
Nominal Voltage	14.8V
Discharge Rate	25C
Maximum Continuous Discharge	5.2 × 25 = 130A
Estimated System Current Draw	35.01A
Maximum Power Delivery	14.8 × 130 = **1924W**



A 25C discharge-rated battery was selected to ensure sufficient current threshold above the estimated 35A system draw. Operating below the maximum discharge capability reduces voltage sag, improves flight stability, and enhances battery reliability during rapid throttle changes.

1.6.3 Endurance Estimation:

$$\textit{Flight Time (Endurance)} = (\textit{Battery Capacity (Ah)} / \textit{Current Drawn by System}) \times 60$$

$$\boldsymbol{Flight\ Time\ (min) = (5.2)/(30.01) \times 60}$$

$$(\boldsymbol{Flight\ Time \approx 10.43\ min})$$

Assuming a 80% maximum battery discharge ≈ $\boldsymbol{8.32\ minutes}$

The endurance estimation is based on the average hover and mission operating current, rather than the peak current consumption, as the maximum current draw is only during short duration manoeuvres such as take-off and fast throttle corrections.

1.7 Material selection:

Material selection is an important aspect in UAVs design due to its direct impact on the structural integrity, vibration characteristics, payload capacity, long term operation, ease of manufacturing and overall flight efficiency. The material selection process was oriented towards finding an optimal compromise between strength, stiffness, impact resistance, vibration damping, and manufacturability.

1.7.1 Material selection criteria

A clear selection criterion and thought process had to be defined for each sub-component.

UAV Subsystem / Component	Primary Selection Criteria
Rotor Arms	High stiffness-to-weight ratio, low structural flex
Arm Holders & Structural Joints	Structural strength and dimensional stability
Landing Gear Interfaces	Impact resistance and energy absorption
Damping Tube Between Arm Holder and Arm Rod	Vibration damping and flexibility
Sensor & Camera Mounts	Vibration isolation and lightweight structure
External Holders & Mounts	Weather resistance and manufacturability
Payload Housing	Lightweight structure and ease of fabrication

1.7.2 Component-wise Material Allocation

Based on these criteria, Carbon Fibre, PETG, and TPU were selected as the primary materials for different UAV subsystems.

UAV Component	Selected Material	Reason for Selection
Rotor Arms	**Carbon Fibre Tube**	High stiffness-to-weight ratio, low flex, and excellent vibration resistance
Central Hub & Mounts	**Carbon Fibre Sheets + PETG**	Good dimensional stability, impact resistance, and easy manufacturability
Landing Gear Interfaces	**TPU + PETG + Carbon Fibre Tube**	Improved impact absorption and vibration damping during landing
Sensor & Camera Mounts	**PETG + TPU**	Flexible structure helps isolate high-frequency vibrations
Payload Mechanism Housing	**PETG**	Lightweight structure with sufficient rigidity and easy 3D printing

1.7.3 Justification for Selected Materials

3K Carbon Fibre – CF Sheets and tubing was selected for the central plates & rotor arms because of its high stiffness-to-weight ratio and low structural flex under dynamic loading conditions. Compared to aluminium and fibreglass, carbon fibre provides significantly lower weight while maintaining high rigidity, improving flight stability and payload efficiency for the proposed autonomous UAV platform.

Parameter	3K Carbon Fibre	Aluminium 6061	Fibreglass Composite
Weight	5	2	3
Structural Rigidity	5	4	3
Manufacturability	3	5	3
Cost	2	3	4

PETG was selected for structural mounts, Joints, Holders and payload housing because it provides a balanced combination of rigidity, impact resistance, print reliability, and manufacturing simplicity. Compared to PLA and ABS, PETG offers better layer adhesion and dimensional stability while remaining easier to print than carbon nylon composites.

Parameter	PETG	PLA	ABS	Carbon Nylon
Rigidity	4	5	3	5
Impact Resistance	4	1	4	5
Printability	5	5	3	1
Cost	4	5	4	2

Although Carbon Nylon offers higher strength and rigidity, PETG was selected because it provides significantly easier printability, lower manufacturing cost, and more reliable dimensional consistency for rapid prototyping and subsystem fabrication.

TPU was selected for damping interfaces and sensor mounts because of its flexibility and vibration absorption capability. Compared to EVA foam and rigid polymer mounts, TPU provides better durability, dimensional consistency, and long-term vibration damping while maintaining sufficient structural support for repeated landing impacts and autonomous flight operations.

The final material selection was optimized to balance lightweight construction, structural reliability, vibration isolation, and manufacturability while meeting the operational requirements of the proposed autonomous UAV platform.

1.8 Subsystem Selection:

1.8.1 Communication System

1) Transmitter & Receiver



The communication subsystem is based on the **RadioMaster TX12 Mark II transmitter** and the **RP1 ExpressLRS 2.4GHz Nano Receiver**, providing reliable long-range and low-latency control. We chose ExpressLRS for its fast packet refresh rate, good resistance to interference and stable signal performance in autonomous and manual operations. The small size of the RP1 receiver also reduces the weight on board while guaranteeing full compatibility with the Pixhawk flight controller and the latest telemetry systems.

Parameter	Specification
Communication Protocol	ExpressLRS (ELRS)
Operating Frequency	2.4 GHz
Refresh Rate	Up to 1000 Hz
Telemetry Support	Bidirectional Telemetry
Receiver Weight	~2.2 g
Control Range	>10 km (ideal conditions)



2) Wi-Fi Connectivity



The onboard Raspberry Pi was connected to the ground station via a **TP-Link 600 Mbps Wi-Fi USB Adapter** for wireless communication. The module provides live video streaming. Wi-Fi communication was used mainly for onboard computing and video streaming while critical flight control was still done by the Pixhawk flight controller for better operational reliability.

3) On-Board Telemetry



The UAV is equipped with the **Simplify SiK Telemetry Radio (433 MHz, 100 mW)** that is directly connected to the Pixhawk flight controller to transmit real-time flight data to the ground station. The telemetry link enables the monitoring of critical parameters such as battery voltage, flight status, GPS data and sensor feedback from the Time-of-Flight and Optical Flow sensors during autonomous operation. The module uses the MAVLink communication protocol and is fully compatible with Mission Planner and QGroundControl for live monitoring and failsafe supervision.

1.8.2 Control System

1) Flight Controller

The **Pixhawk 2.4.8** flight controller was selected because it provides stable and reliable low-level flight control while remaining lightweight and well-supported for autonomous UAV development. Unlike high-performance racing flight controllers such as the SpeedyBee F405, the Pixhawk supports advanced autonomous features including GPS navigation, MAVLink telemetry, waypoint missions, and seamless integration with Mission Planner and Raspberry Pi-based onboard computing.

The Pixhawk Cube series provides better sensor redundancy and processing capability, but it was not selected due to its significantly higher cost, larger form factor, and unnecessary computational overhead for the proposed mission profile. Since high-level vision processing and payload logic are handled separately by the Raspberry Pi, the Pixhawk 2.4.8 provides sufficient processing capability for stable flight control, navigation, and failsafe operations while maintaining lower system complexity and weight.

Feature	Specification
Processor	STM32F427VIT6
Input Voltage	4.1V – 5.7V
Dimensions	81.5 mm × 50 mm × 15.5 mm
Weight	~38 g
Communication Support	MAVLink, UART, I2C, PWM



Parameter	Pixhawk 2.4.8	Pixhawk Cube	SpeedyBee F405
Affordability	8	4	9
Autonomous Compatibility	9	10	5
Documentation & Community Support	10	8	6
Sensor & Peripheral Integration	9	10	6
Weight Efficiency	8	6	9
Overall Suitability	**44**	38	35

2) Companion Computer

Raspberry Pi 5 Model B (8GB RAM) is the onboard computer for high-level autonomous operations, onboard vision processing, and real-time sensor fusion. Autonomous flight includes QR detection, camera processing, Time-of-Flight sensor integration, Optical Flow

processing and payload related decision making. The onboard computer is also linked to the Wi-Fi telemetry for live video streaming and real-time data transmission to the ground station.

The Raspberry Pi 5 provides much better CPU performance and faster I/O communication than the Raspberry Pi 4, which is critical for parallel camera feeds and sensor fusion tasks with lower latency in processing. The board communicates to the Pixhawk flight controller via UART MAVLink. Low level stabilization and flight critical tasks are still handled by the Pixhawk.

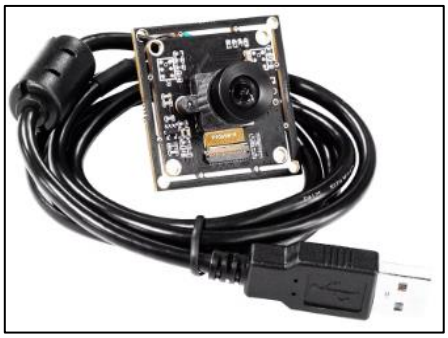
We considered the NVIDIA Jetson Nano, which has stronger GPU acceleration for AI workloads. But it was not selected due to its higher power consumption, higher thermal requirements and increased overall system cost. The Raspberry Pi 5 is a more balanced solution in terms of processing capability, power efficiency, integration simplicity, and weight, since the proposed mission mainly requires lightweight OpenCV processing, QR-based navigation, and moderate real-time vision computation instead of heavy neural network inference.

Feature	Specification
Processor	Quad-Core ARM Cortex-A76
RAM	8GB LPDDR4X
GPU	VideoCore VII GPU
Wireless Connectivity	Dual-Band Wi-Fi + Bluetooth 5.0
Communication Interfaces	UART, USB 3.0, CSI, GPIO
Operating System	Raspberry Pi OS (Linux Based)



1.8.3 Navigation & Avionics

1) Camera Module 1: The Arducam Global Shutter Colour USB Camera Board



The **Arducam Global Shutter Colour USB Camera** was selected for the forward-looking vision sensor for the autonomous navigation, obstacle awareness, colour detection and real-time environmental sensing. Global shutter cameras acquire the whole frame at the same time in contrast to rolling shutter cameras and therefore relieves motion distortion during fast drone movement and quick attitude change. This enhances the robustness of object detection and frame processing towards autonomous flight.

The camera was selected based on its high frame rate capability, low-latency USB interface, and ability to acquire stable images during dynamic motion conditions. Its small footprint and native USB support for the Raspberry Pi 5 also make it easy to integrate for real-time OpenCV-based processing and onboard sensor fusion applications.

2) Camera Module 2: The Arducam IMX708 Camera Module (12MP)

The downward facing camera for QR detection, payload targeting, landing alignment and visual ground tracking was selected to be the **Arducam IMX708 12MP Camera Module**.

During low-altitude autonomous operations, the camera continually observes the landing and payload drop region.



The IMX708 was primarily chosen for its high resolution imaging and wide field of view to increase the accuracy of QR detection and target visibility during descent. Higher image detail allows better object localization and payload alignment at different altitudes than low resolution camera modules. The CSI interface also allows for faster image transfer and lower processing latency than the standard USB webcams, making it more suitable for real-time precision-based autonomous tasks.

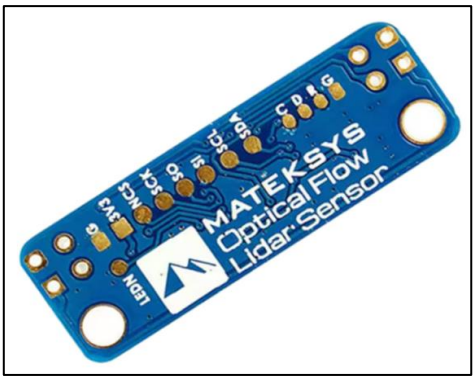
3) GPS Module: Neo M10N GPS Module



The **Holybro M10 GPS** module was selected for real-time positioning, waypoint navigation, Return-to-Home (RTH), and autonomous mission tracking. The module supports multi-constellation GNSS reception, improving position accuracy and signal reliability during outdoor autonomous operations.

The Holybro M10 was specifically selected because of its reliable compatibility with the Pixhawk 2.4.8 flight controller and native MAVLink-based ecosystem support.

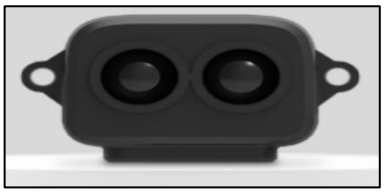
4) Optical Flow Sensor



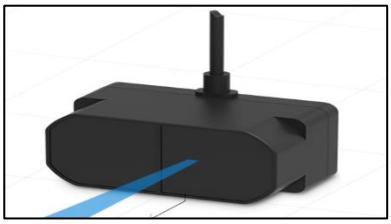
The **Matek 3901-L0X Optical Flow & LiDAR sensor** was selected for low-altitude position holding and accurate height estimation during autonomous corridor navigation. The optical flow sensor tracks ground movement while the integrated LiDAR measures vertical distance from the ground surface, helping the UAV maintain stable hover and controlled motion in GPS-limited environments.

This module is particularly useful during the low-altitude obstacle corridor section and payload approach phase, where precise altitude control and stable movement are critical.

5) Time of Flight Sensors



The **Benewake TFmini** Plus is used as the front-facing LiDAR for wall-distance measurement and obstacle detection during corridor navigation. Its narrow 3.6° FoV and high refresh rate improve accurate front-distance sensing and camera-based obstacle avoidance.



The **Benewake TF-Luna** is used as the downward-facing LiDAR for altitude estimation and terrain-relative height holding along with the Optical Flow sensor. It was selected because of its lightweight design, stable short-range ranging, and low power consumption for continuous low-altitude operation.

1.9 Autonomous Navigation System Hardware & Software

The sensor architecture, interconnections, communication protocols and data flow between the Pixhawk flight controller, the Raspberry Pi 5 companion computer, the onboard sensors and the ground dashboard for the UAV system.

1.9.1 Core System Architecture

The UAV is based on a dual-computing architecture:
• Pixhawk 2.4.8 handles low-level stabilisation, flight control, GPS navigation, failsafe, and motor control.
• Raspberry Pi 5 handles high-level autonomous operation including OpenCV processing, QR detection, object tracking, obstacle analysis, and payload logic.
• Communication between Raspberry Pi and Pixhawk is done using MAVLink over UART.
• Ground station communication via 433 MHz telemetry and Wi-Fi streaming.

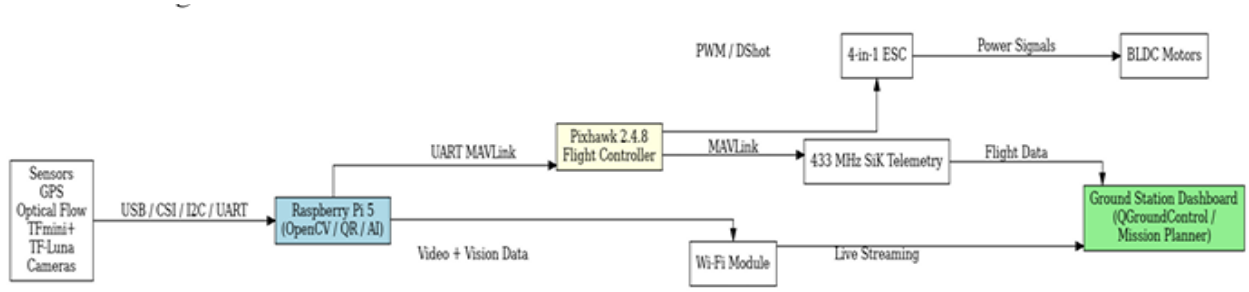


Fig. System Architecture

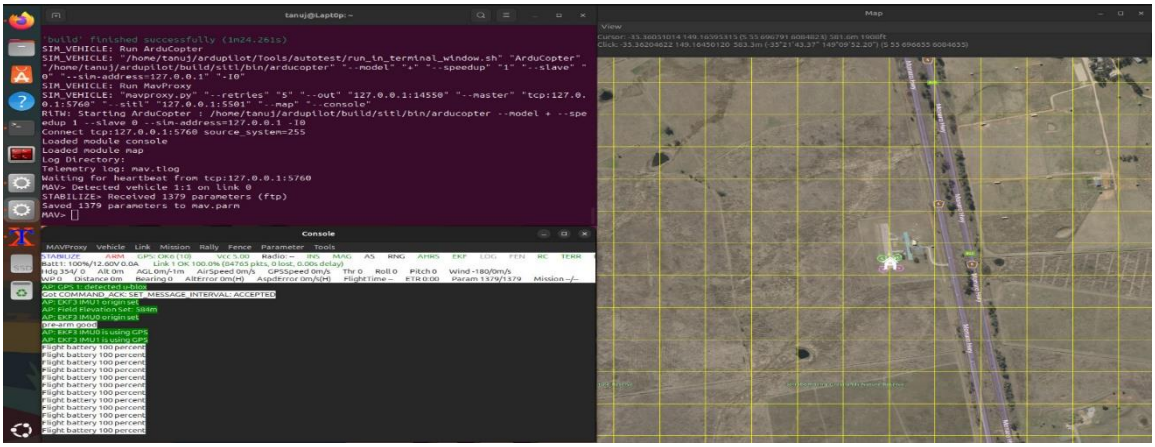


Fig. Autonomous navigation software validation environment using ArduPilot SITL, MAVProxy, and simulated GPS-based navigation.

1.9.2 Sensors and Their Connections

Sensor	Purpose	Connected to	Interface	Data Type
Arducam Global Shutter Camera	Forward vision and obstacle sensing	Raspberry Pi 5	USB	Video frame
Arducam IMX708 Camera	QR detection and landing alignment	Raspberry Pi 5	CSI	High resolution image

Neo M10N GPS	Positioning and waypoint navigation	Pixhawk	UART	GNSS coordinates
Matek Optical Flow	Low altitude stabilization	Pixhawk	12C/UART	Motion vector
TFmini Plus	Front obstacle distance sensing	Raspberry Pi 5	UART	Distance measurement
TF-Luna	Altitude estimation	Pixhawk	UART/12C	Height data

1.9.3 Detailed Data Flow

1.Cameras and LiDAR sensors constantly collect information about the surroundings.

2. Raspberry Pi performs image processing with OpenCV and QR detection algorithms.

3. Raspberry Pi combines data from the camera and distance sensor and sends navigation commands to Pixhawk over MAVLink over UART.

4. The Pixhawk handles IMU, GPS, optical flow and companion computer commands.

5. Pixhawk produces motor outputs through PWM/DShot signals to ESC.

6. The BLDC motors are driven by the ESC for stabilisation and navigation.

7. The SiK telemetry module is used for transmission of the flight telemetry to the ground dashboard.

8. Live video and vision system data are visible to the operator via Wi-Fi streaming**.**



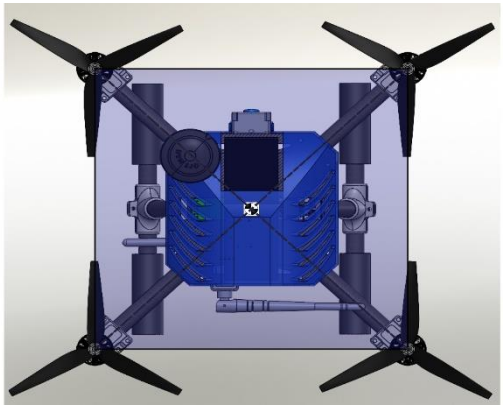
Fig. Data flow

1.9.4 Operational Workflow

During autonomous flight the Raspberry Pi serves as the smart processing layer while pixhawk still handles real-time flight stabilisation. Sensor data from cameras, LiDAR, GPS and optical flow modules are processed continuously to maintain stable navigation, corridor following, QR-code detection, obstacle avoidance and payload deployment.

1.10 CG Calculation and stability

The Center of Gravity (CG) is one of the most important parameters in the design of a UAV. The CG directly influences flight stability, maneuverability, payload carrying capability and control response. The CG analysis of the UAV was carried out in the horizontal (X-Y) and vertical (Z) planes.



1.10.1 Horizontal Plan (X-Y) axis

The horizontal CG analysis was performed to verify the lateral and longitudinal balance of the UAV. The center of gravity was placed at a position near the geometric center of the four motor mounts, this guaranteed a symmetric mass distribution over the frame.

X - direction

$$X_{CG} = \frac{\Sigma (m_i \times x_i)}{\Sigma m_i}$$

Y – direction

$$Y_{CG} = \frac{\Sigma (m_i \times y_i)}{\Sigma m_i}$$

* Calculation :–

$m_1 = 50$, $x_1 = 279.2$, $y_1 = 279.2$

$m_2 = 50$, $x_2 = -279.2$, $y_2 = -279.2$

$$X_{CG} = \frac{(50 \times 279.2) + (50 \times (-279.2))}{50 + 50}$$

$X_{CG} = 0$

$$Y_{CG} = \frac{(50 \times 279.2) + (50 \times (-279.2))}{50 + 50}$$

$Y_{CG} = 0$

Mass properties of Drone with Mechanism
Configuration: Default
Coordinate system: -- default --

Mass = 1649.48 grams

Volume = 7670033.53 cubic millimeters

Surface area = -163647447.41 square millimeters

Center of mass: (millimeters)
X = 678.82
Y = 931.45
Z = 1352.51

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)
Taken at the center of mass.
Ix = (-0.19, -0.29, 0.94) Px = 15745328.70
Iy = (-0.61, -0.71, -0.35) Py = 16252594.15
Iz = (0.77, -0.64, -0.04) Pz = 16810301.41

Moments of inertia: (grams * square millimeters)
Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)
Lxx = 16563225.58 Lxy = 301988.79 Lxz = -73830.26
Lyx = 301988.79 Lyy = 16437218.81 Lyz = -152678.53
Lzx = -73830.26 Lzy = -152678.53 Lzz = 15807779.88

Moments of inertia: (grams * square millimeters)
Taken at the output coordinate system. (Using positive tensor notation.)
Ixx = 4465026523.47 Ixy = 1043247610.06 Ixz = 1514345360.81
Iyx = 1043247610.06 Iyy = 3793908616.05 Iyz = 2077853265.45
Izx = 1514345360.81 Izy = 2077853265.45 Izz = 2206965557.50

One or more components have overridden mass properties:
Raspberry Pi 4 Model B.step
esc.STEP
TPU holder<1><Default>
Pixhawk 2.4.8<1><Default>
FS-IA10B 2.4Ghz<1><Default>
TPU holder<2><Default>

1.10.2 Vertical Plane (Z Plan)

- The vertical position of CG plays a significant role in the natural stability characteristics of the UAV, especially during transportation and payload release.
- The UAV is designed with an underslung payload configuration that maintains the overall centre of gravity beneath the plane of propeller thrust. This causes a stabilising pendulum effect which improves self-righting behaviour and decreases pitch and roll oscillations in flight.

1.10.3 CG Calculation with payload:

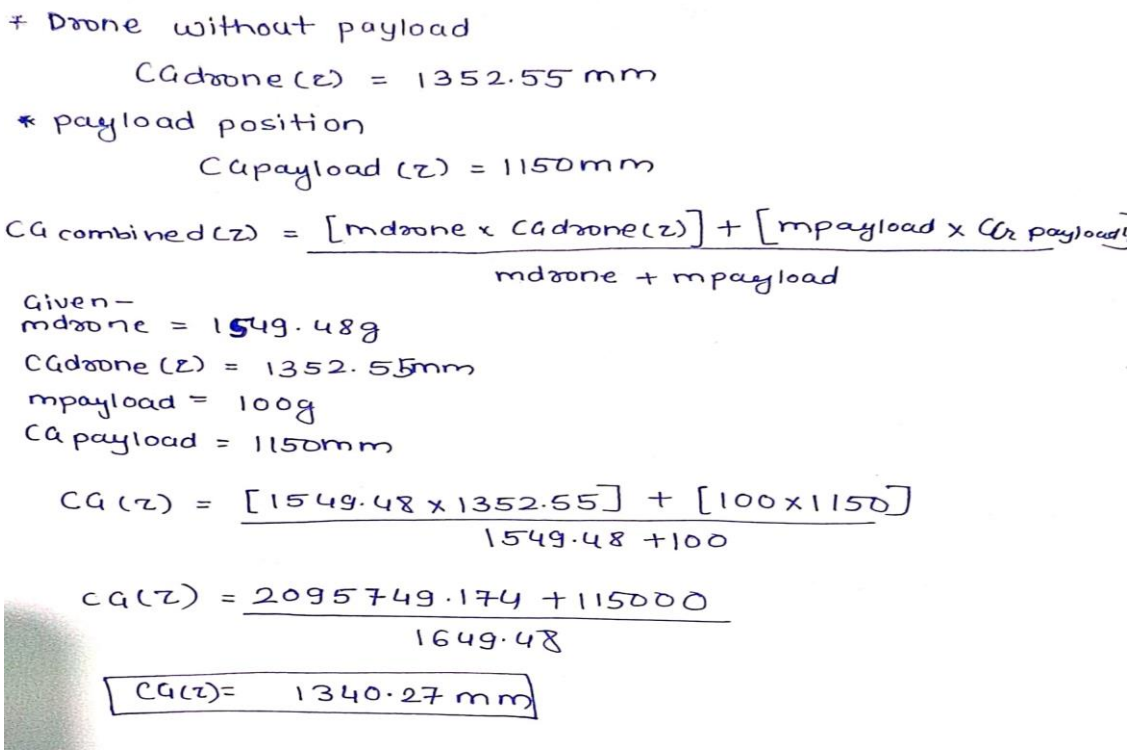
Axis	Value
X	678.82
Y	931.81
Z	1352.51

CG with payload= (**678.62, 931.81, 1352.51**) mm
Drone mass=**1649.48 gram**

1.10.4 CG Calculation without payload:

Axis	Value
X	678.69
Y	921.91
Z	1352.55

CG without payload= (**678.69, 921.91, 1352.55**) mm
Drone mass=**1549.48 gram**



2.0 Computational Analysis

2.1 Ansys Structural & Vibration Analysis

Computational analysis was conducted in **ANSYS** to support the iterative design development and validate the structural performance of the proposed UAV. The simulations helped the design decisions by studying the effect of geometry, choice of material and structural modifications during the development process. Structural, vibrational and impact analyses were performed on critical components to evaluate stress distribution, deformation, resonance behaviour and landing load resistance under expected operating conditions.

2.1.1 Rotor Arm Static Structural Analysis

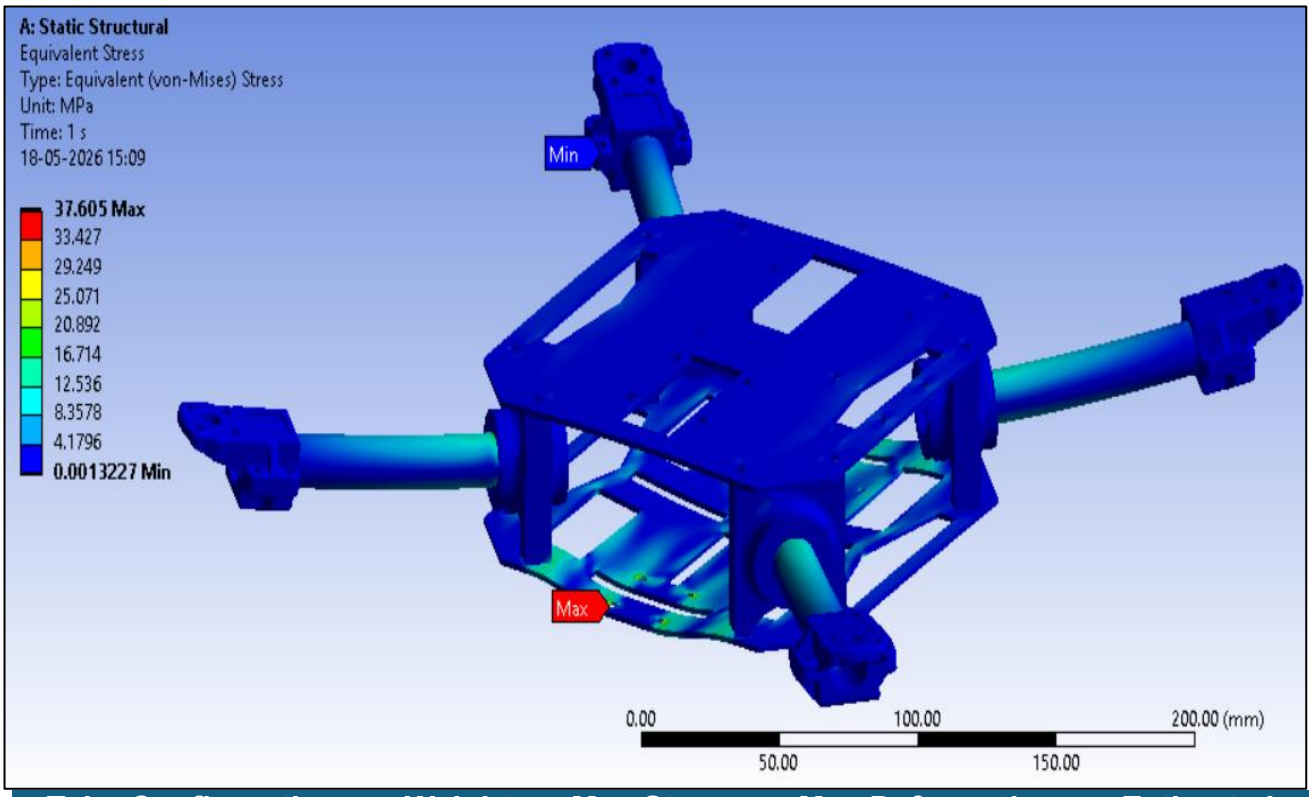
Objective: To optimize the rotor arm tube dimensions by balancing structural rigidity, factor of safety, and overall system weight.

Methodology

Parameter	Value
Analysis Type	**Static Structural**
Fixed Support	Landing Gear Screws
Applied Load	20 N Upward Force (Per Arm)
Total Applied Load	80 N
Mesh Size	2 mm Global Mesh
Average Element Quality	0.833
Average Skewness	0.224

Four carbon fibre tube configurations with different outer diameter (OD) and inner diameter (ID) combinations were evaluated under identical loading and boundary conditions using static structural analysis.

The analysis also indicates that most of the structural load is carried by the carbon fibre plates and rotor arm tubes, while comparatively lower stress is observed in the arm-holder interfaces due to effective load distribution across the frame structure. This ensures that the vulnerable components stay protected.



Tube Configuration (OD-ID)	Weight (g)	Max Stress (MPa)	Max Deformation (mm)	Estimated FoS
20–18 mm	12	24.8	0.384	9.3
18–16 mm	10	30.9	0.521	7.4
16–14 mm	8	37.6	0.712	**6.1**
14–12 mm	6	52.4	1.186	4.4

As the motor's maximum thrust is 1760g per motor. A 20N (almost 2 kg) force was applied on each motor in the upward direction against gravity.

Based on the structural optimization study, the **16–14 mm** carbon fibre tube configuration was selected as it offered the best compromise between structural rigidity, deformation control, and weight reduction, while providing a safe factor of safety under aggressive flight loading conditions. The optimized design resulted in approximately 16 g weight reduction for all four rotor arms compared to the original design of 20–18 mm with no significant compromise on the structural performance.

2.1.2 Modal Analysis

The modal analysis in ANSYS was used to study the vibration behaviour of the UAV frame during flight. The propulsion system is constantly creating vibrations that can influence the flight controller, cameras and onboard sensors.

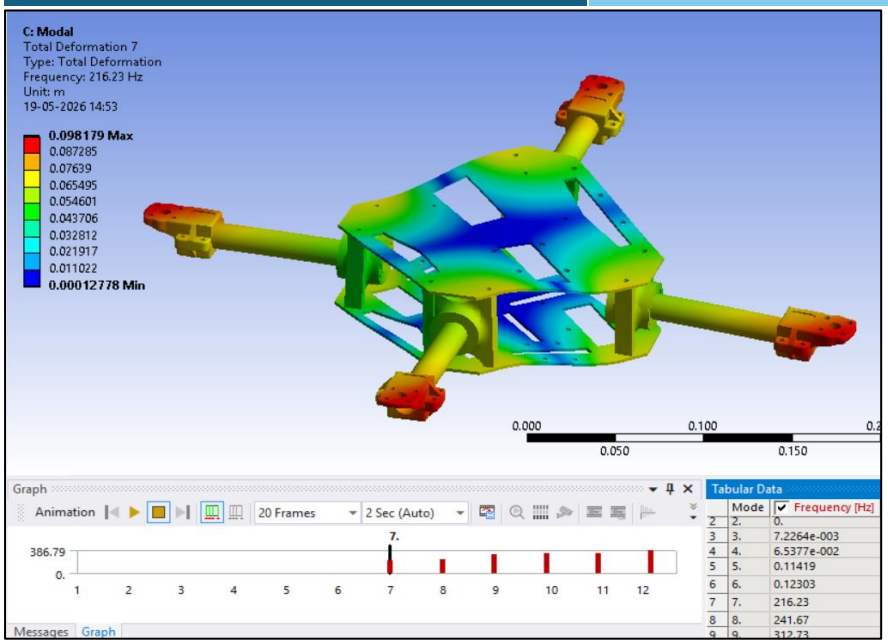
According to the motor datasheet, the operational excitation frequency for the expected thrust range of 500-800 g (13200 – 15000 rpm) was estimated as follows:

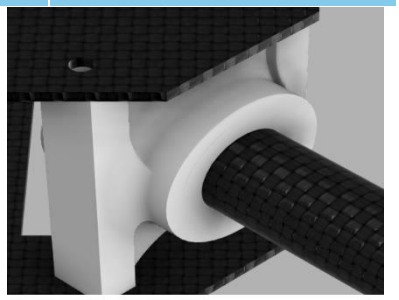
For the operational RPM range:

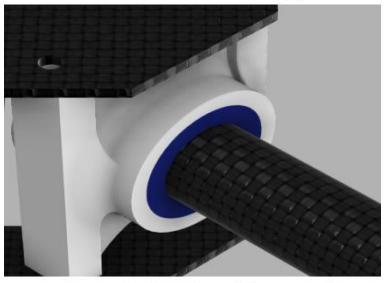
$$= \frac{13200-15000}{60}$$
$$\approx 220-250\ Hz$$

Initially, the frame was analysed using a rigid arm-holder configuration without any damping material. The first structural modes were found close to the operational motor vibration range, indicating possible vibration transfer from the rotor arms to the main frame and resonance effects. To reduce this, a **TPU** isolation layer was introduced between the carbon fibre arm and the arm holder.

Parameter	Rigid Holder	TPU Holder
Mode 7 Frequency	216.23 Hz	204.46 Hz
Mode 8 Frequency	241.67 Hz	230.26 Hz
Mode 9 Frequency	312.73 Hz	281.86 Hz




Initial Holder Design


TPU Optimized Holder Design

The optimized TPU-based arm-holder design also reduced approximately 13–14 g of overall structural weight compared to the initial rigid-holder configuration.

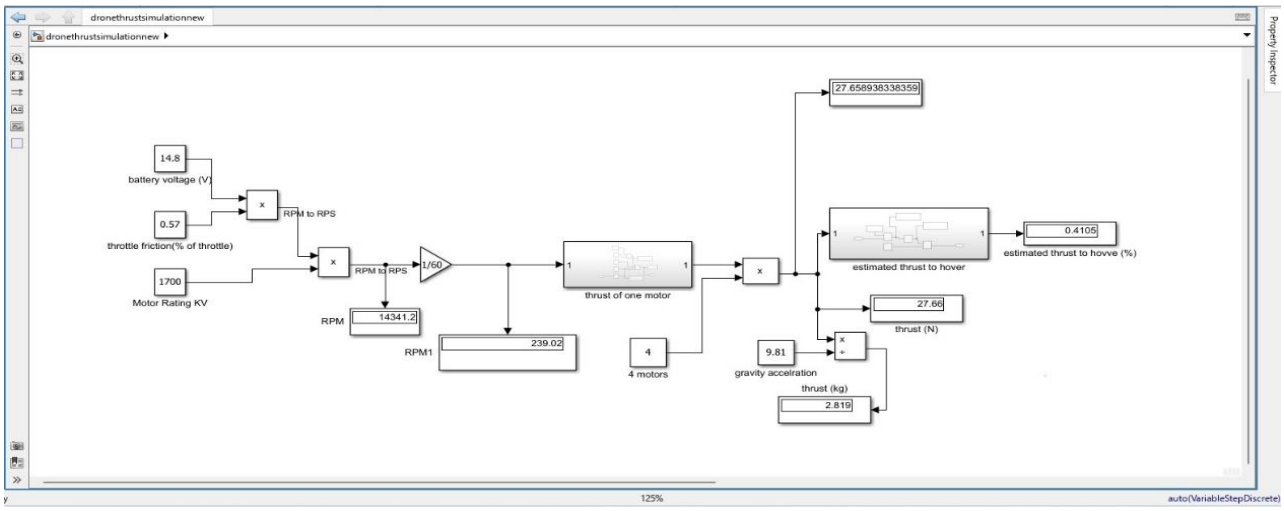
2.2 MATLAB Analysis

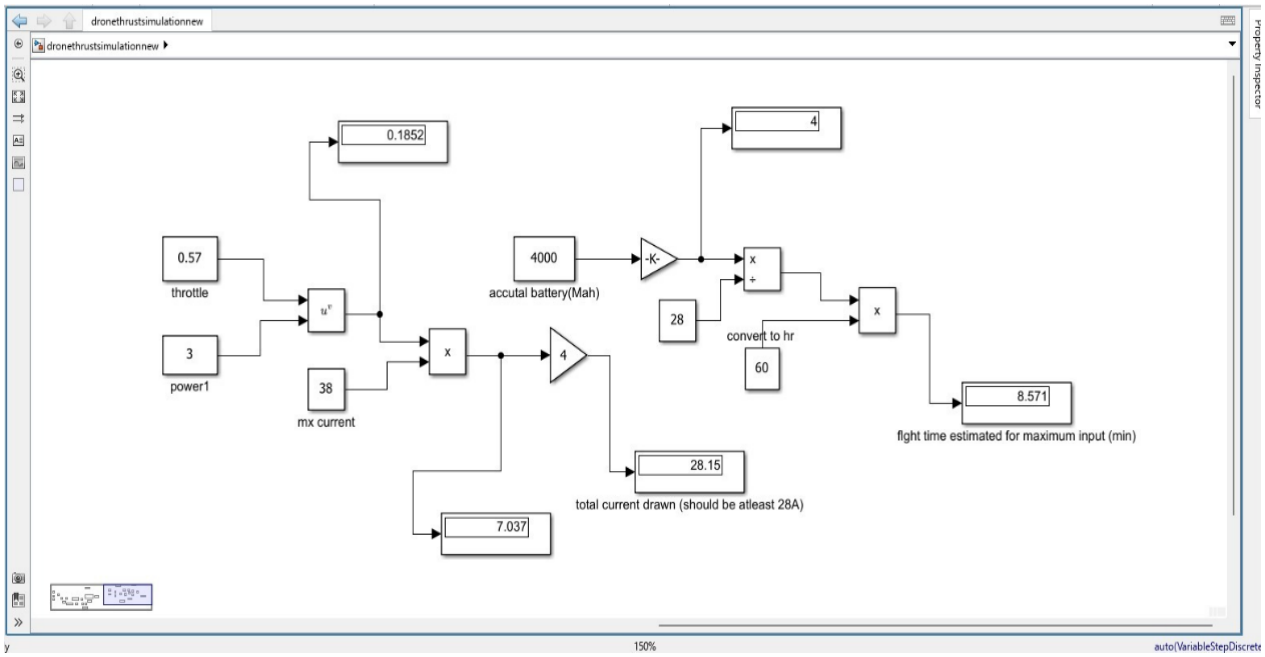
2.2.1 Thrust-to-Weight & Power Estimation Simulation

The simulation of thrust and weight was done in **MATLAB Simulink**, where the flight time and current consumption during flight was estimated. It is powered by a 4S battery and an EMAX ECO II Series

2207 1700KV brushless motor. The propulsion system is a 7x4x3 tri-blade propeller. The model predicts the motor current draw from the throttle coefficient using a cubic relationship of the throttle input to motor current. For the calculations a maximum current of 38A per motor was considered. This results in a total current consumption of approximately 28.15A for all four motors, which corresponds to a current draw of almost 7A per motor. The simulation gave an estimated flight time of about 8 minutes and 57 seconds.

Considering the drone's hover dynamics, the generated thrust should be equal to the total weight of the drone plus wind drag and other resistive forces. The total thrust of ~2.8kg was calculated based on the input voltage, throttle input and motor KV rating of the simulation. This model was useful to understand the dynamics of the drone and to estimate the overall performance characteristics of the quadcopter system.





2.2.2 PID Flight Stability Simulation

This is a MATLAB Simulink model of a closed loop PID based attitude control system for a quadcopter (drone). The three sections correspond to the **three motions** of rotation of the drone:

- Yaw
- Roll

- Pitch

In each loop the desired angle command (reference input) is compared with the actual output angle through a feedback connection, generating an error signal. This error is passed through the **PID controller** structure where the signal is divided into three control actions,

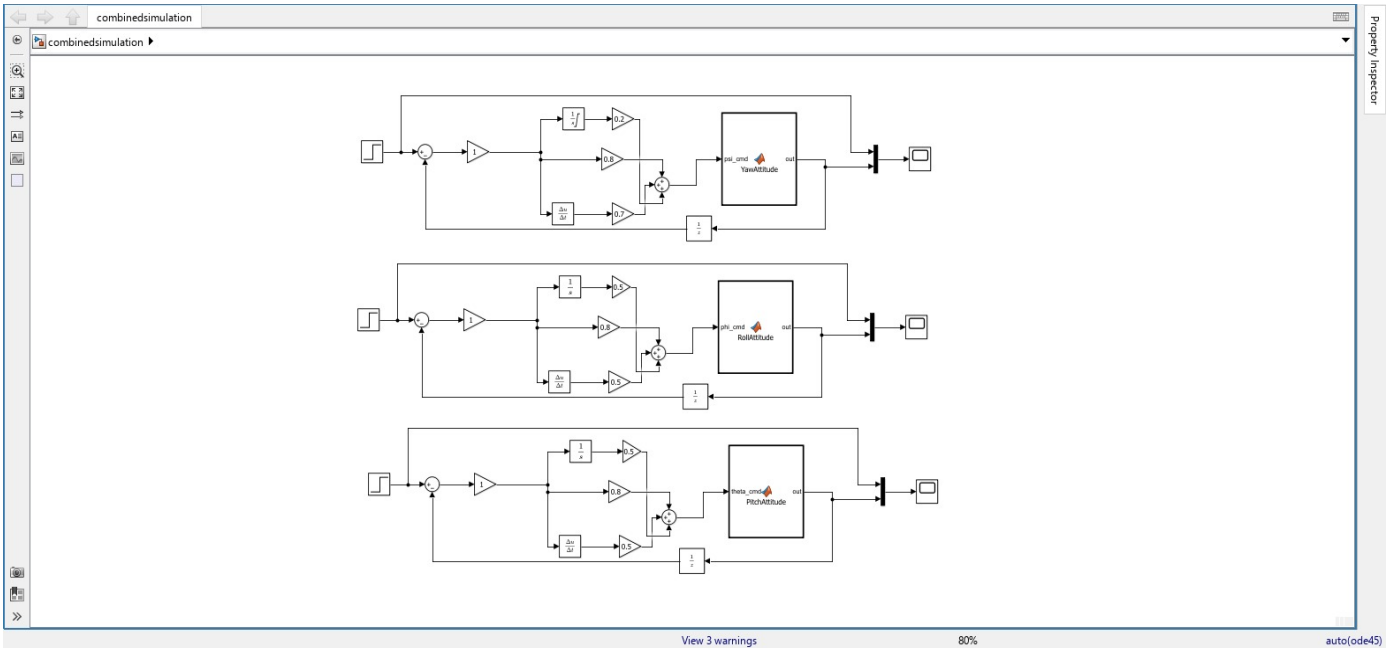
- Proportional(P)
- Derivative (D)
- Integral (I)

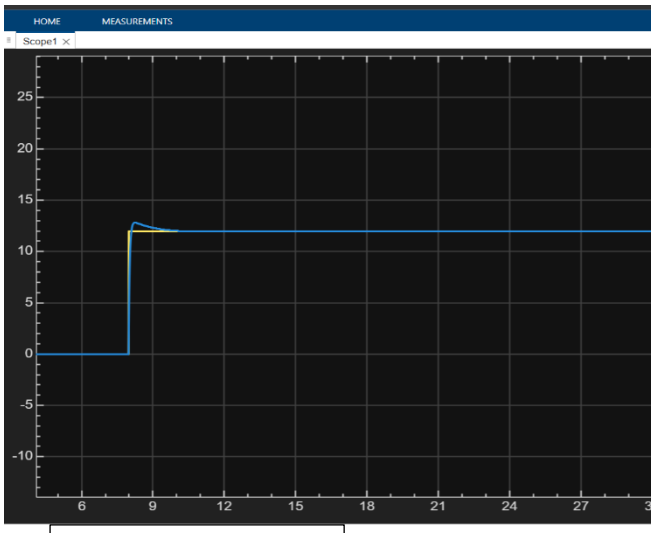
The proportional path responds immediately to the present error and accelerates the response, the integral path accumulates previous errors to remove the steady-state offset, and the derivative path predicts the upcoming error tendencies to decrease oscillations and enhance stability. The gains adjacent to each branch (e.g. 0.2, 0.5, 0.8, etc.) determine the magnitude of each PID component and are directly related to the system performance such as rise time, overshoot and settling time.

Then the output of the three PID action is added together and the control signal is sent to the subsystem blocks named as:

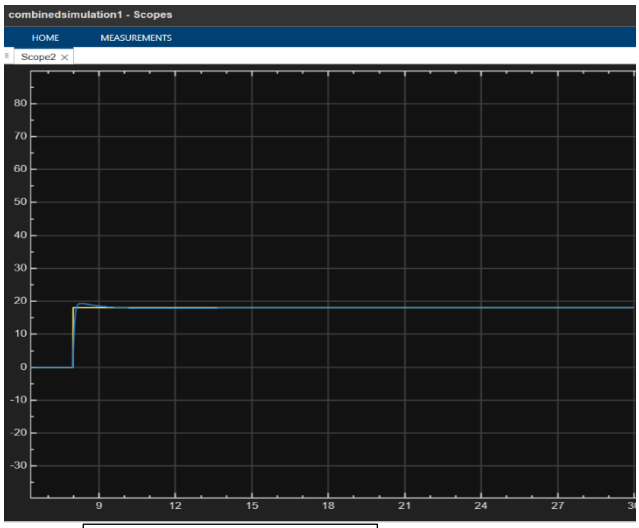
- Yaw attitude
- Roll attitude
- Pitch attitude

which represent the mathematical dynamics of the drone for each axis. The system output is continuously monitored using scopes, while the feedback loop returns the measured attitude back to the summing junction for correction. This continuous correction mechanism allows the drone to maintain stable orientation even when disturbances or command changes occur. The model is designed separately for each axis because yaw, roll, and pitch have different dynamic behaviors and tuning requirements. Overall, this architecture demonstrates how PID control stabilizes a quadcopter by minimizing attitude error and ensuring smooth, accurate rotational motion in all three axes.

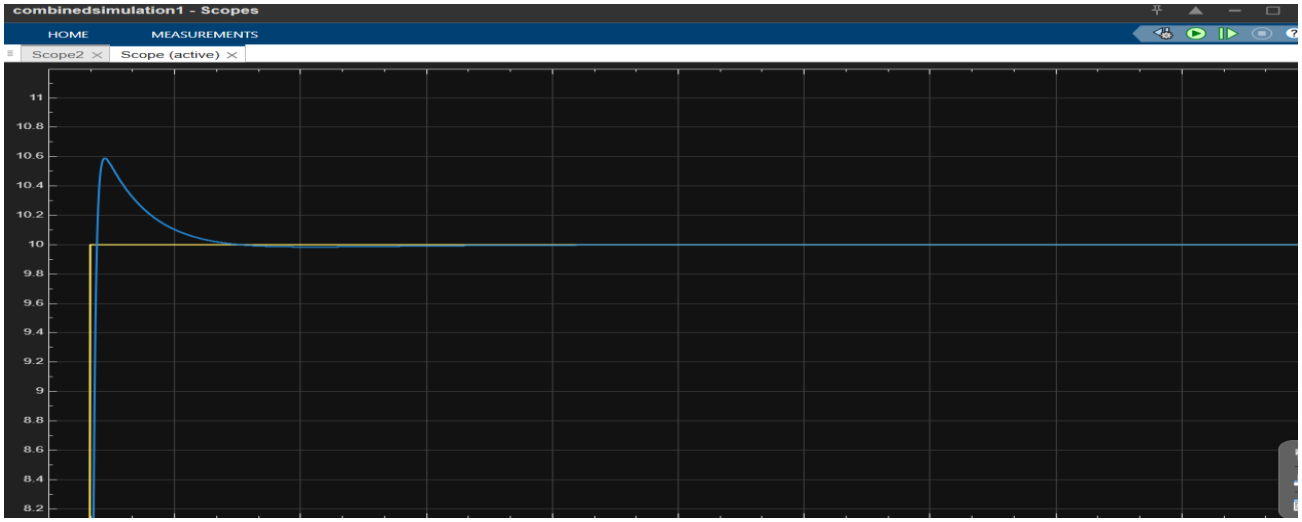




For Yaw



For Roll



For Pitch

3.0 Methodology for Autonomous Operation

3.1 Autonomous Flight Algorithm

3.1.1 Autonomous Navigation Methodology



Fig. Opencv + Pyzbar QR detection with center alignment
(movement instructions are generated in real time for autonomous navigation)

Autonomous Navigation Methodology The autonomous navigation system is developed using **Python, OpenCV, Pyzbar, and MAVLink** communication. The drone first performs system initialization, GPS verification, and autonomous take-off using Pixhawk flight control commands. QR code detection is carried out using a downward-facing camera with image pre-processing and multi-frame confirmation for reliable target identification.

For corridor entry and navigation, the UAV uses HSV colour masking to detect the green entry banner and aligns itself automatically before entering the corridor. Inside the corridor, OpenCV-based Canny edge detection and Hough Transform techniques are used to estimate the corridor centre and maintain stable navigation using visual correction methods.

After exiting the corridor, the drone performs waypoint-based safe-zone search while avoiding restricted areas. Multiple QR codes are scanned and matched with the stored target QR information. The UAV then performs precision target locking using visual alignment before deploying the payload through a servo-controlled winch mechanism. Finally, the drone automatically initiates Return-To-Launch (RTL) for safe mission completion.

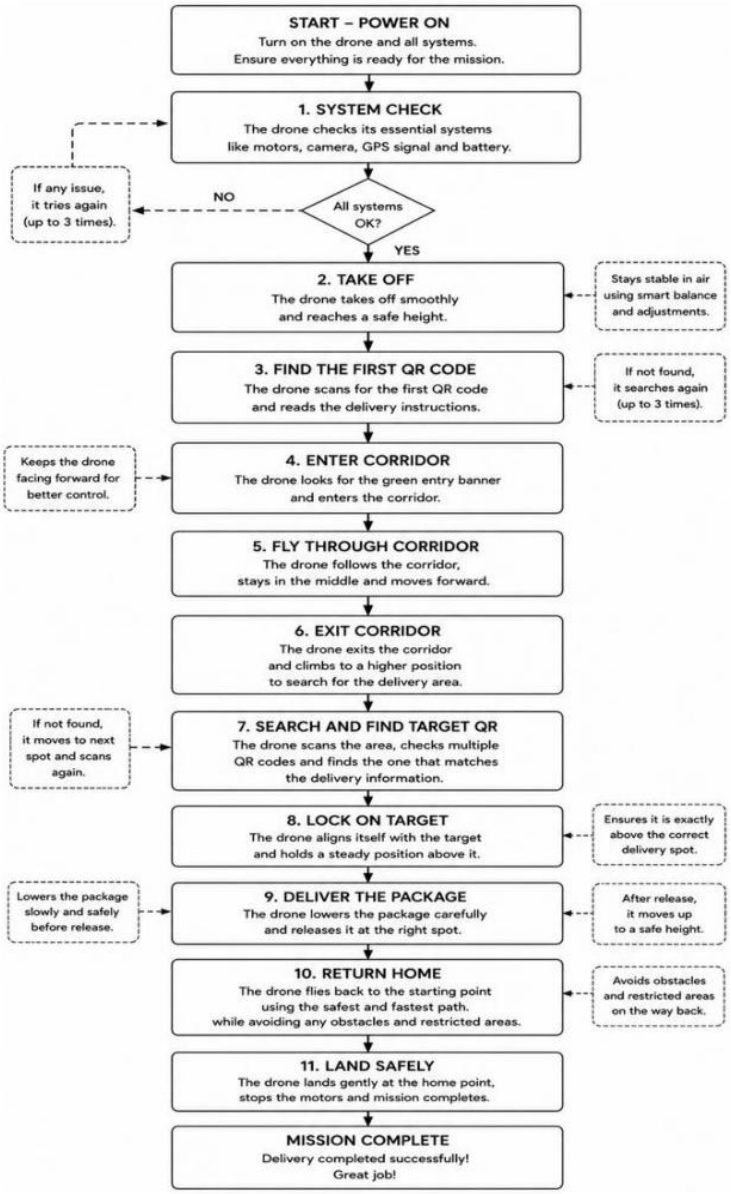


Fig. Autonomous Mission Flowchart

3.1.2 OpenCV + Pyzbar QR Detection – centre alignment & movement units

The QR code is detected using OpenCV and pyzbar in real-time. The green bounding box shows the detected QR code, the red dot represents the centre of drone camera frame, while green dot represents the centre of QR code bounding box.

The system calculates the offset and provides movement instructions to align the QR code with camera centre for precise navigation.

Key points:

- Red dot: Drone camera centre
- Green dot: QR code BBox centre
- Movement units: Relative distance in pixels

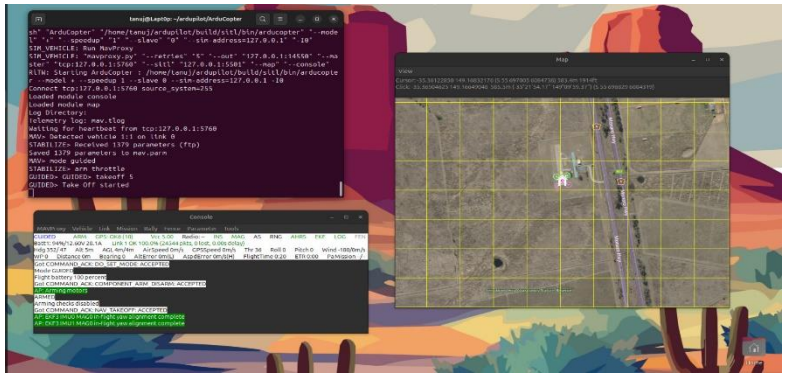


Fig. Autonomous takeoff and stable hover validation in GUIDED mode using MAVLink-based flight commands in ArduPilot SITL.

3.2 Autonomous object detection and counting Algorithms

Our object detection strategy is built around QR codes. Each target in the safe zone carries a unique QR code; the drone must read it, match it against the code scanned at take-off, then centre itself over the match before deploying the winch. All vision processing runs locally on the Raspberry Pi 4B no ground-station assistance, no cloud inference.



Fig. Real-time QR code detection and decoding for autonomous target identification using OpenCV and pyzbar.

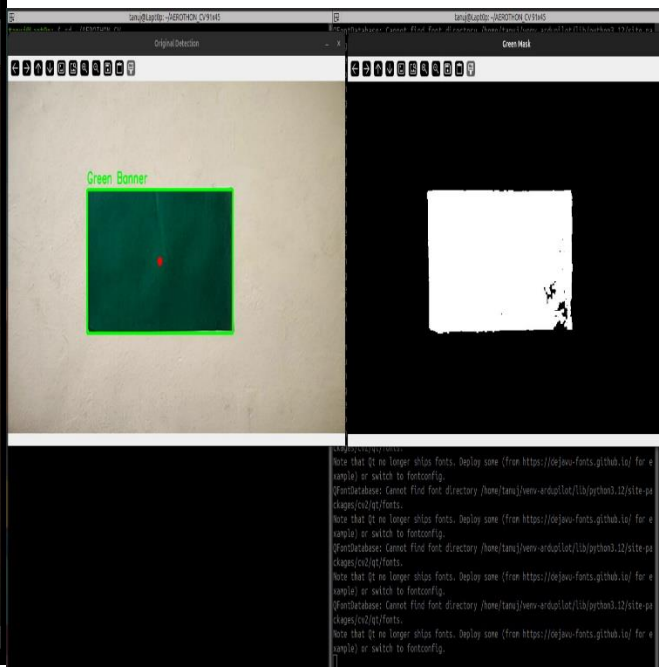


Fig. HSV-based green banner detection and centroid localization for autonomous corridor recognition.

3.2.1 Detection Pipeline Overview

Every frame from the down-facing camera passes through the following seven stages before a detection is either confirmed or discarded. Stages 1-3 run on every frame; stages 4-7 only run when pyzbar returns at least one candidate:

Step	Stage	Processing operation	Output/decision
1	Frame capture	OpenCV Video Capture at 30 fps, native res	Raw BGR frame ready for processing
2	Pre-processing	Resize to 640 x 480, then convert to grayscale	Grayscale ndarray (half the pixel counts faster decode)
3	QR decode	pyzbar. decode () - C-library scan of the entire grayscale frame	List of decoded objects, each with data string and bounding polygon
4	Validation	Track the last three decoded strings; accept only if all three matches	Confirmed QR string or REJECT (treated as noise)
5	Matching	Compare confirmed string against the QR stored in State 3 memory	MATCH proceed to centroid or NO MATCH → continue search

6	Centroid extraction	Compute centroid (cx, cy) from the four-corner bounding polygon	Pixel coordinates passed to the visual servo
7	Visual servo	pixel_error = (cx-W/2, cy-H/2); P-controller maps error to velocity	Drone centres within 15 px tolerance → winch enable flag set

3.2.2 Corridor Navigation - Visual Processing

While flying through the corridor (State 5), the front camera feeds a separate vision thread that keeps the drone centred between the two walls. We chose edge detection over colour tracking here because the corridor walls are likely to be a neutral colour under variable lighting - edges are more robust in those condition.

3.2.3 Object Counting and QR Inventory

As the drone scans the safe zone during the lawnmower search pattern, every detected QR code is catalogued. The table below describes the counting and deduplication logic;

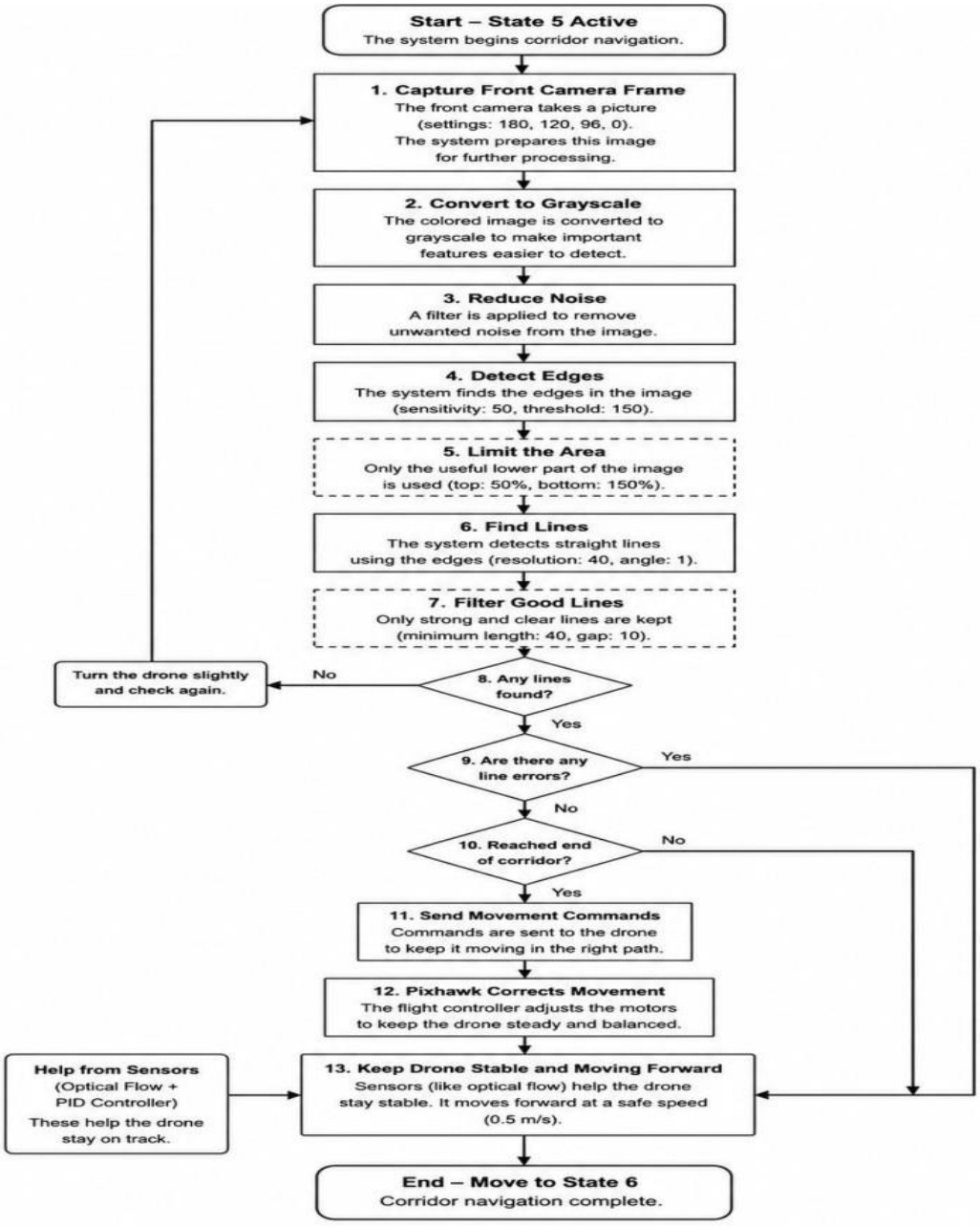
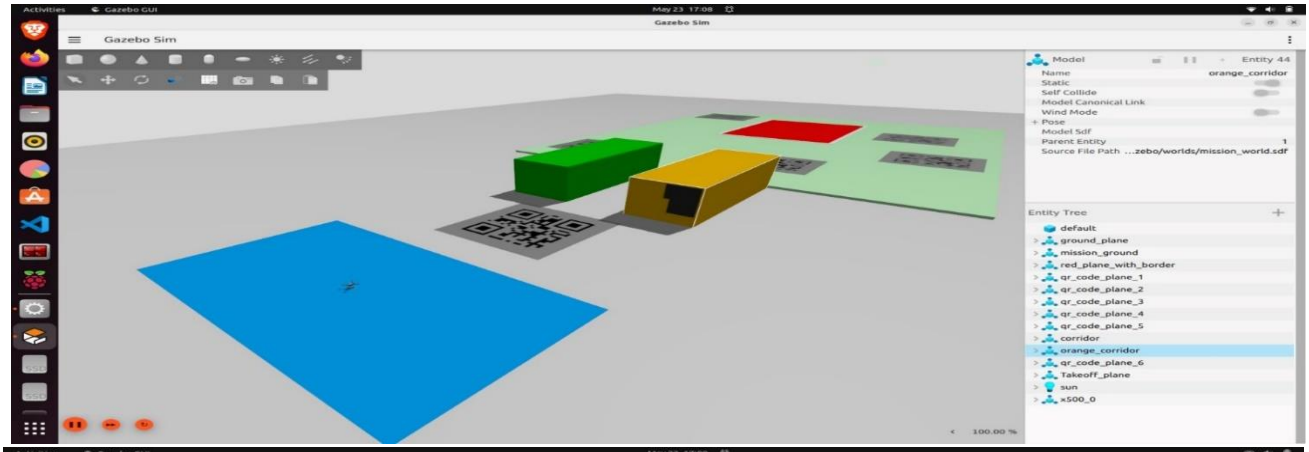


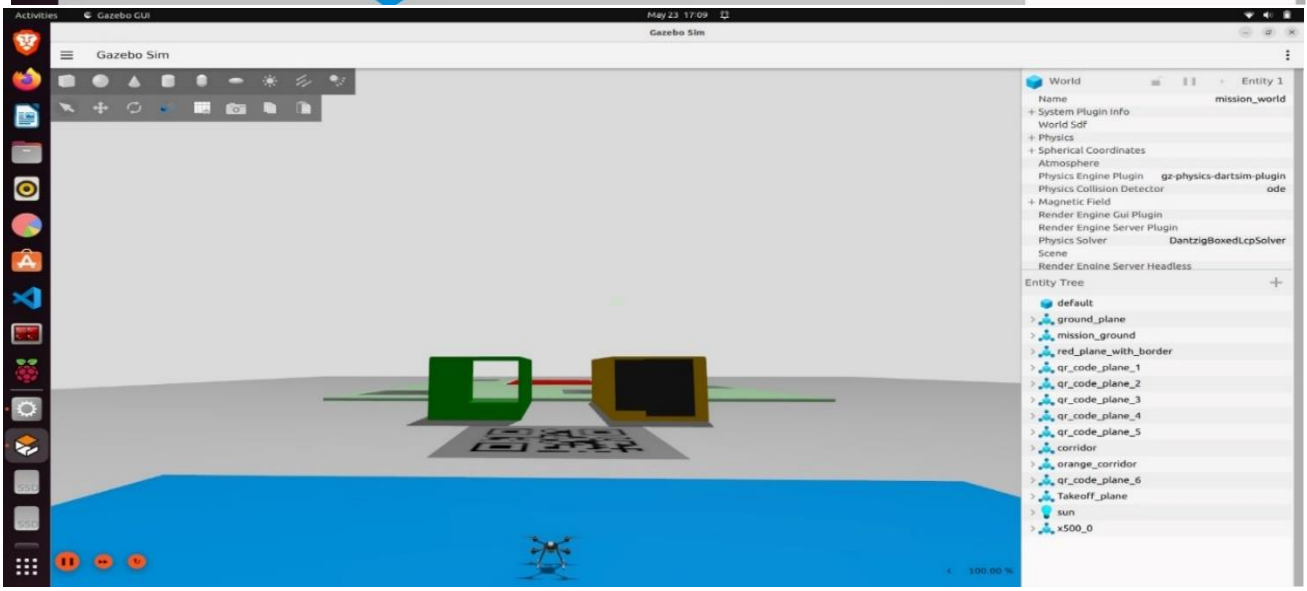
Fig. Corridor Navigation

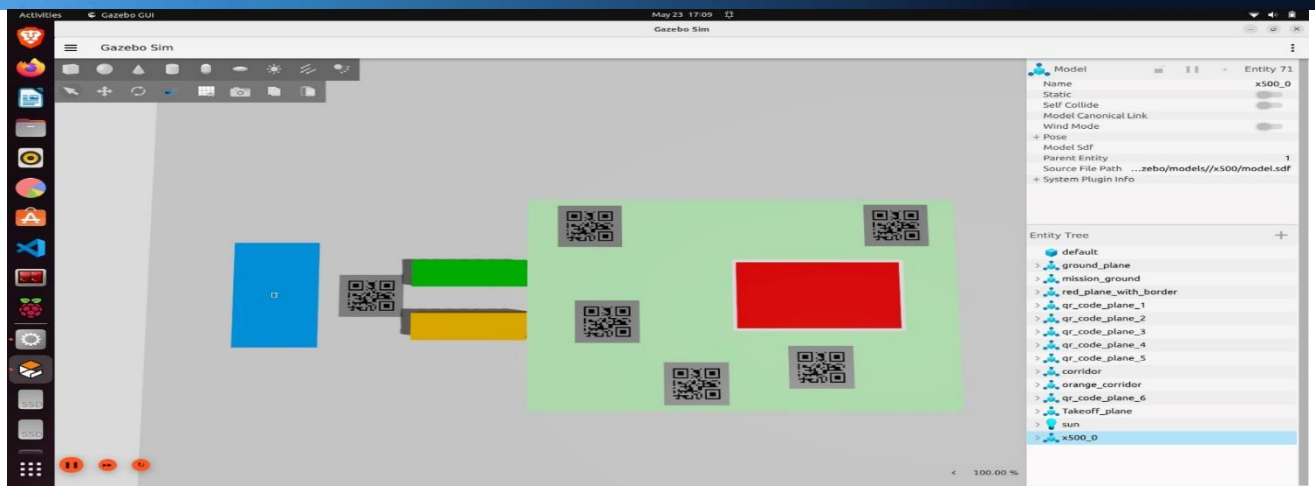
Function	Implementation	Purpose
QR inventory	Python dict {qr_string: (lat, lon, timestamp)}	tores every unique QR code seen, with its GPS location and time
Deduplication	Key-based lookup before any insertion into the dict	Ensures the same QR code is never counted twice regardless of how many times it is scanned
3-frame confirmation	Decoded string must match across three consecutive frames	Eliminates false positives from partial reads or motion blur
GPS tagging	MAVLink GLOBAL_POSITION_INT appended at detection time	Geo-references each detection so we can navigate back to a match
Match check	Dict keys compared against stored starting-QR string	Immediately triggers transition to Precision Lock (State 9) on a match
Log entry	CSV row: timestamp, QR data, lat, lon, alt	Provides a complete audit trail of all objects detected during the mission

3.2.4 Detection Performance Parameters

We tested the pipeline at multiple altitudes and under indoor fluorescent and outdoor overcast lighting before settling on the values below. The 3-frame confirmation threshold was the single biggest lever for reducing false positives:







3.3 Autonomous Payload Drop Mechanisms

3.3.1 Autonomous system Architecture

Our system splits responsibilities cleanly between two boards. The Raspberry Pi 4B runs all mission-level logic in Python - it decides where to go, what to look for, and when to act. The Pixhawk takes care of the low-level work: sensor fusion, attitude estimation, and motor mixing. The two boards talk to each other over a serial/USB link using the MAVLink protocol, accessed from Python via the pymavlink library.

This split keep concerns separate. If our QR-detection code takes a few extra milliseconds, it does not stall the stabilisation loop - the Pixhawk carries on regardless. That isolation is the main reason we chose this architecture over running everything on a single board.

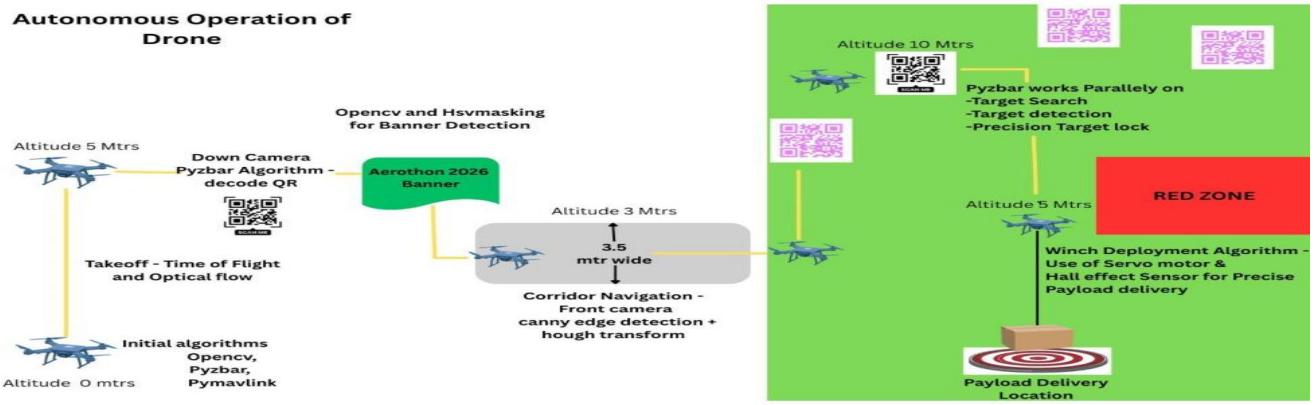


Fig. Autonomous Operation of Drone

3.3.2 Mission Planner - Full Flight Path

The mission is planned using QGroundControl (QGC) mission planner. The complete flight path including take-off, corridor entry, delivery point and return to home is shown in the map. The side panel displays real-time telemetry information such as altitude, speed, battery status, distance, heading and flight mode.

The planned mission path includes all major stages of the operation such as:

- Autonomous take off from the launch point
- Navigation towards the QR scanning area
- Entry into the designated corridor
- Autonomous traversal through the corridor
- Movement towards the delivery zone
- Payload delivery at the target location
- Safe return to the home position

KEY POINTS:

- Total Waypoints: 11
- Mission Type: Autonomous
- Flight mode: Auto
- Take off Altitude: 10m AGL
- Return mode: RTL (return to launch)

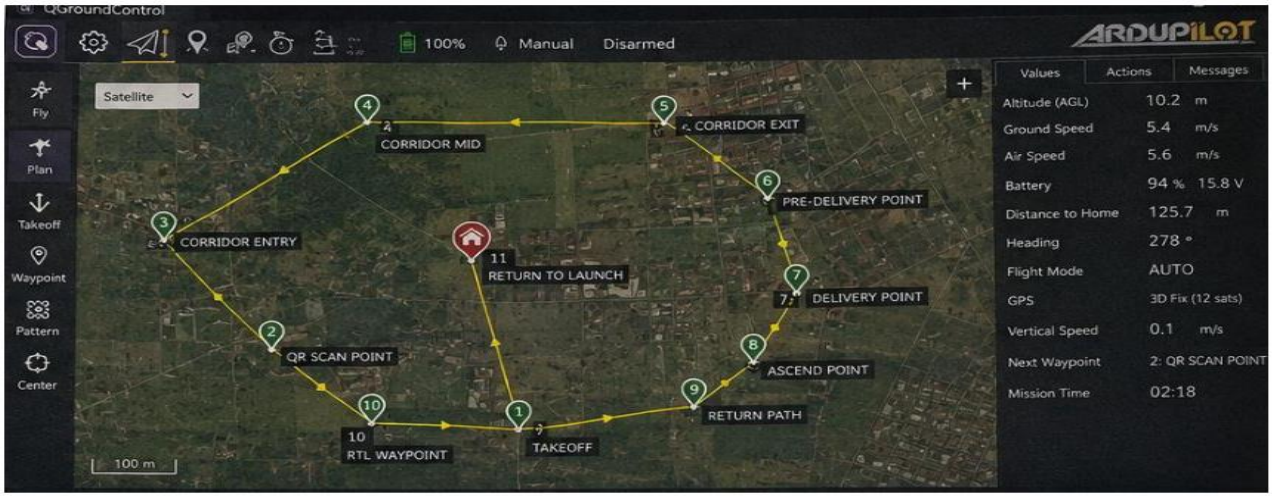


Fig. mission planner showing complete flight path with waypoints & real time telemetry.

3.3.3 Mission summary

Sr.no	State	Camera	Key method	Exit Condition
1	Initialization	Both	System health check	All systems OK
2	Take off	-	MAVLink NAV TAKEOFF	Alt ≥ 4.75 m
3	QR identification	Down	pyzbar + 3-frame confirm	QR stored in memory
4	Corridor Entry	Front	HSV green banner detection	Aligned + entered
5	Corridor navigation	Front	Canny + Hough centreline	Corridor end detected
6	Exit corridor	-	MAVLink altitude set 10 m	Alt ≥ 9.5 m
7	Target detection	Down	Multi-QR scan + matching	QR matches stored code
8	Winch deployment	-	Hall sensor pulse count	String ≥ ground dist + slack
9	Payload release	-	Slack detect + RTL command	LAND mode confirmed

3.3.4 Dashboard

The flight dashboard provides real-time monitoring and control of the autonomous drone mission. It displays important flight parameters such as GPS status, altitude, speed, battery health, optical flow data, and TOF sensor readings during operation. The dashboard also shows live camera feed, QR decoding status, and sensor health monitoring to ensure stable and safe autonomous navigation throughout the mission.



4.0 Flight Safety

Flight safety is an integral part of the UAV design, integrated in structural, electronic and operational subsystems. The system has been designed in line with the design philosophy to handle failures gracefully, minimising risk to personnel, equipment and the operational environment.

4.0.1 Fail Safe Configuration

- Return to Launch on Low Battery - A two-stage voltage monitoring system is implemented, with a warning alert activated at 14.0V (3.5V per cell), and an automatic RTL trigger at 13.6V (3.4V per cell).

- Propeller & Payload Safety - Nylon 7x4x3 tri blades flex on impact and do not shatter making it much safer around obstacles.
  All props are checked for cracks or balance problems before each flight and fitted with self-locking motor nuts
  The MG90 servo payload release is spring-loaded to the hold position by default.

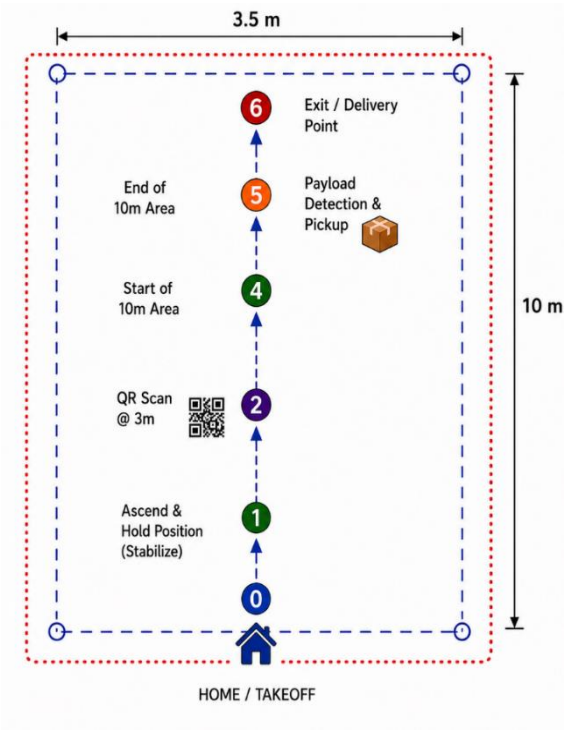
- Geo-Fence – A software boundary driven by Pixhawk and Holybro M10 GPS keeps the UAV within the boundary and notifies the operator before any autonomous action.
  A must-have for autonomous operations.

- Arming Safety - Motors will only spin when both hardware arming switch and throttle low confirmation are met.
  Pixhawk does a pre-arm check before arming. This covers GPS lock, compass calibration, sensor health and RC link.
  Eliminates any chance of accidental motor activation while handling and setting up.
- Structural Safety
  Each of the arm holder joints has TPU isolation layers to isolate propulsion vibrations (220-250Hz) from the main frame and avionics.
  ANSYS confirmed a FoS of 6.1 on the selected 16-14 mm carbon fibre rotor arms at 20 N per arm, well above the design target FoS > 4.

4.1 Safety Feature & SORA Assessment

4.1.1 Geofencing

For safety of drone and the mission to be executed properly, we are using Geofencing in which the Drone will function and complete its task in a specific region. Geofencing can be used in our mission to restrict the drone movement inside the predefined 3.5m × 10m operational area during autonomous navigation and payload handling. If the drone drifts outside the allowed boundary due to wind or instability, the system automatically performs path correction, Return-To-Home, or emergency landing operations.

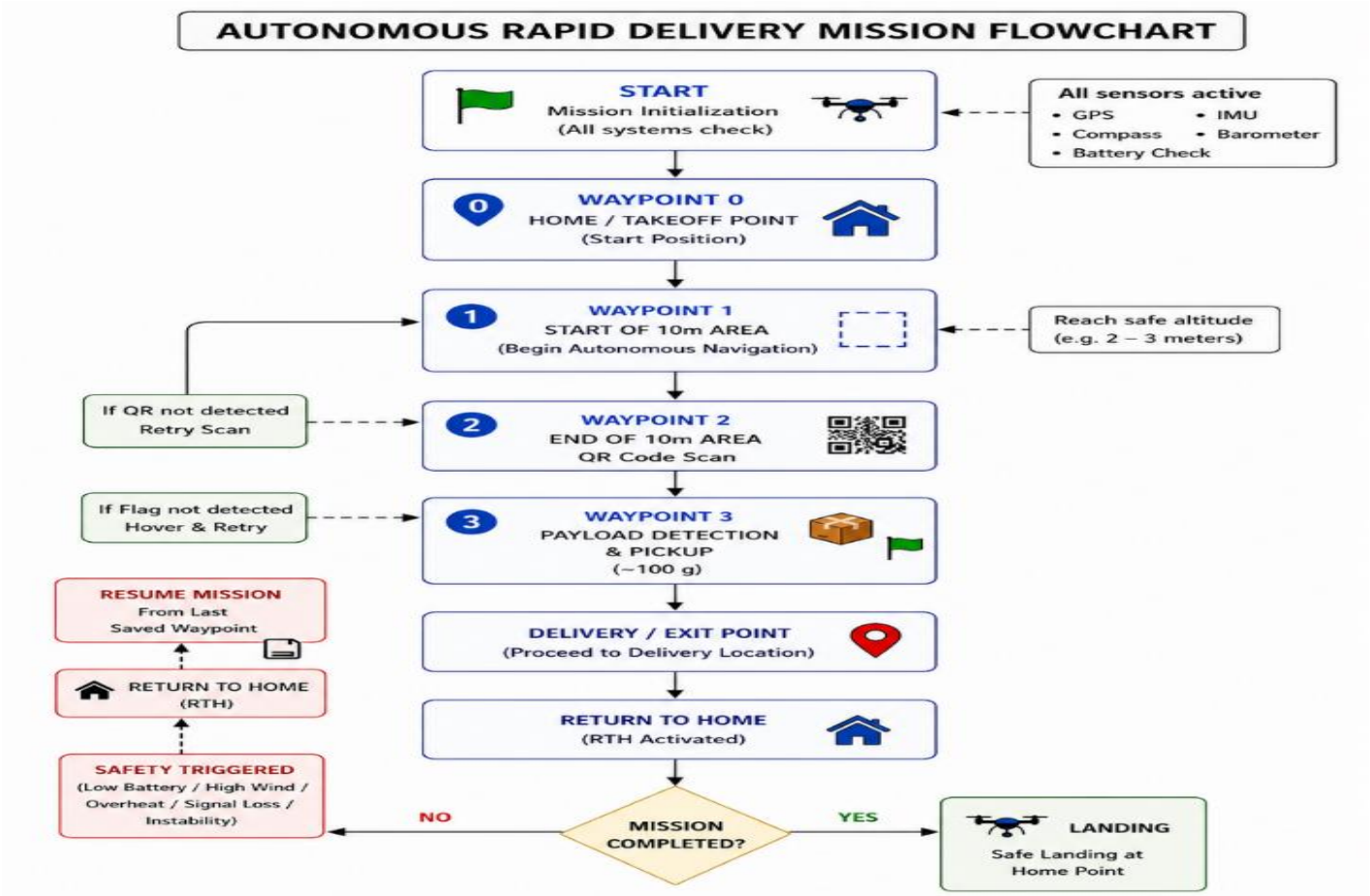


4.1.2 Return to Home and Path Correction

We have used the Return-To-Home (RTH) system to safely bring the drone back to the starting position whenever unstable conditions such as low battery, high wind speed, motor overheating, or signal loss are detected during the mission.

We have added a path correction mechanism in the system to maintain stable navigation inside the predefined operational area during autonomous flight. If the drone starts drifting due to wind disturbance, payload imbalance, or positional instability, the flight controller automatically detects the deviation and corrects the path using IMU and navigation feedback. This helps the drone remain inside the geofenced boundary and continue the mission safely without unnecessary interruption

For stability of drone, gyroscope is already present in drone;



We have also added a mission resume mechanism in which the drone stores the last completed waypoint before returning home, so that after recovery the mission can continue from the same point instead of restarting from the beginning.

Thus, if any of unavoidable circumstances occurs such as low Battery, Lost GPS signal, Sensor and other electronic components malfunctions, there are possible safety measures taken.

4.1.3 SORA Assessment

In our mission, we have mainly used SORA to identify the possible risks during an autonomous flight and to add proper safety actions before the drone enters into unsafe conditions. Our drone flies inside a limited geofenced area of 3.5 m × 10 m, scans the QR code at 3 m, detects the green flag, picks a payload of around 100 g, and safely returns. The overall mission risk is controlled.

The main risks of the ground are: payload drop, drone drift, crossing the boundary, unstable landing, collision with objects during autonomous movement. To avoid such risks, we have added geofencing, path correction, Return-To-Home, emergency landing and mission resume from last saved waypoint.

The risk for the air is quite small since the mission is conducted in a small region and at a low height. But we have considered wind disturbance, GPS error, signal loss, battery overheating and motor heating as important safety conditions. In case of any unsafe condition, the drone aborts the mission, stores the last waypoint and return to home and resumes from the same waypoint after the recovery.

In simple terms, SORA helps us to demonstrate that our drone mission is planned with proper safety logic, risk control and recovery operation and not just delivering the payload.

4.1.4 Emergency landing

We have added an Emergency Landing System in the drone to safely land the drone whenever critical unstable conditions such as motor overheating, excessive tilt, or signal loss are detected during the mission. Instead of continuing unsafe flight, the drone will automatically search for a stable position and perform controlled landing to prevent drone damage, payload loss, and collision risk inside the operational area.

5.0 Innovation

5.0.1Innovation 1: Gripping Mechanism

1) Objective

The gripping mechanism aims at designing a payload delivery system that can be used on UAVs for carrying and accurate releasing of 100 g payload. The system is to be lightweight, reliable and energy efficient. The mechanism is to allow controlled lowering of the payload and automatic release while maintaining the stability and aerodynamic efficiency of the UAV.

The developed system comprises a servo-controlled rope lowering mechanism and gravity based automatic hook release. The fabrication of the structure is done with PETG 3D printed parts and the controlled actuation is performed by an MG90 servo motor. A torsion spring mechanism is used to provide a constant tension on the fishing line and smooth operation for payload deployment and retrieval.

2) Initial Considerations and Comparisons

During the conceptual design phase, different payload gripping and release mechanisms were studied and compared based on parameters such as weight, complexity, power requirement, reliability, aerodynamic effect, and suitability for UAV applications.

The following gripping systems were considered;

Parameter	Hook release mechanisms	Electric gripper	Box/container gripper
Weight	Very lightweight	Medium to heavy	Heavy
Complexity	Simple mechanism	Complex mechanisms	Moderate
Power consumption	Very low	High	Medium
Manufacturing cost	Low	High	Medium
Payload release speed	Fast	Medium	Slow
Maintenance	Easy	Difficult	Moderate
Suitable for 100g payload	Highly suitable	Overdesigned for small payload	Suitable but bulky

3) Advantages

- Lightweight Design
- Low Power Consumption
- Simple and Reliable Operation
- Controlled Payload Lowering
- Improved Aerodynamic Performance
- Easy Fabrication and Maintenance
- Cost-Effective Solution

For this purpose, we used the hook released mechanisms for better stability, fast releasing, simplicity and for weight reduction.

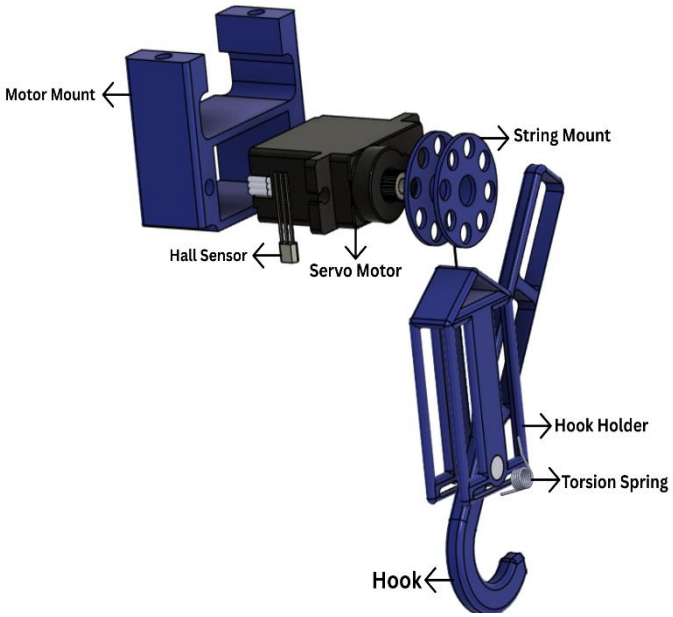
Objective: To develop an energy efficient and lightweight autonomous payload release system.

Requirements:

To develop lightweight mechanism that can be used many times without damage and works automatically. The mechanism should securely hold the payload during operation.

Our Solution:

We are using automatic hook release mechanism it works using servo-controlled rope lowering system and a gravity based automatic release hook. The mechanism is 3D printed from **PETG material** and also **uses servo MG90** for lowering the fishing line and torsion spring to provide constant rotational tension.



Process:

At start the payload is securely attached to the hook, the hook

remains locked because payload weight creates tension. Torsion spring used for maintain proper tension and smooth operation. When drone reaches at target point servo rotates and lowers the payload using fishing line. Instead of directly dropping the payload, the fishing line allows the payload to descend gradually and accurately to the target point, reducing impact and improving safety. The payload release process is also unique because it uses gravity-based counterweight automatic hook. When the payload reaches the ground, the ground surface supporting the payload weight and decreases the tension in fishing line. Due to reduction in tension, the counterweight lever attached to the mechanism becomes unbalanced. Gravity pulls counterweight downward which causes rotational motion about pivot point. This motion shifts the locking lever and automatically opens the hook mechanism. When the hook opens the payload released safely onto the target point.

5.0.2 Innovation 2: Hall Effect Sensor Based Precise Payload Lowering System

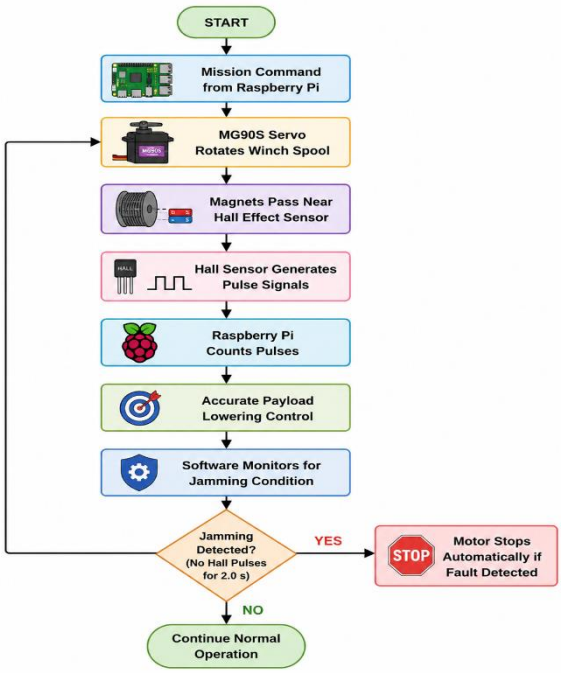
1) Basic Idea

The developed payload lowering system uses a Hall Effect sensor integrated with a servo-driven winch mechanism to achieve accurate and reliable payload deployment in the UAV. Neodymium magnets mounted on the 3D-printed spool generate pulse signals as the spool rotates, which are detected by the Hall Effect sensor and processed by the Raspberry Pi through GPIO interrupts. By counting these pulses, the system precisely determines the amount of tether deployed and accurately controls the payload lowering height.



Unlike conventional time-based deployment systems, this mechanism measures actual physical spool rotation, making it immune to battery voltage fluctuations during flight. The system also incorporates an anti-jamming safety feature, where the Raspberry Pi continuously monitors Hall sensor feedback and automatically stops the motor if spool rotation is not detected, thereby preventing servo stalling, electrical damage, and mechanical failure.

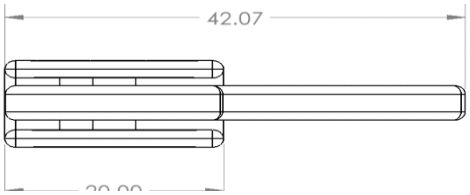
2) Working

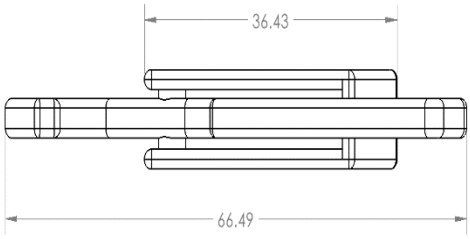


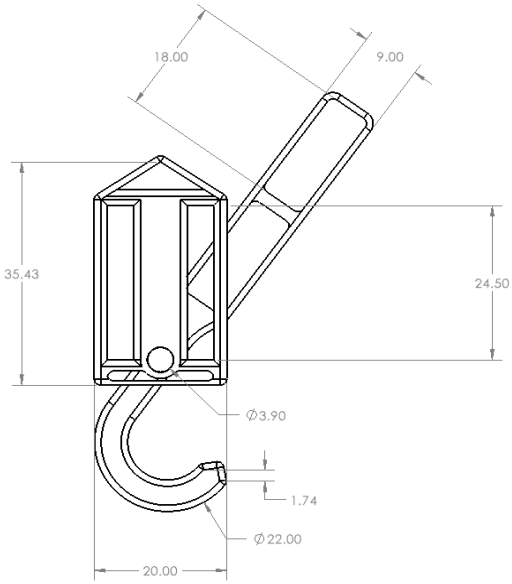
The Hall Effect sensor increases accuracy and reliability of the payload lowering system by measuring the actual rotation of the spool, rather than time of motor operation. Unlike traditional systems, it is insensitive to variations in battery voltage and allows for accurate payload deployment in flight. Magnetic pulse detection of continuous spool movement provides accurate tether control and improved operational stability. The complete Hall Effect sensing assembly – comprising the sensor, magnets, and wiring – adds **only about 2 grams to the UAV's weight, while delivering sophisticated feedback and safety features, making it a highly efficient and lightweight innovation.**

3) Advantages

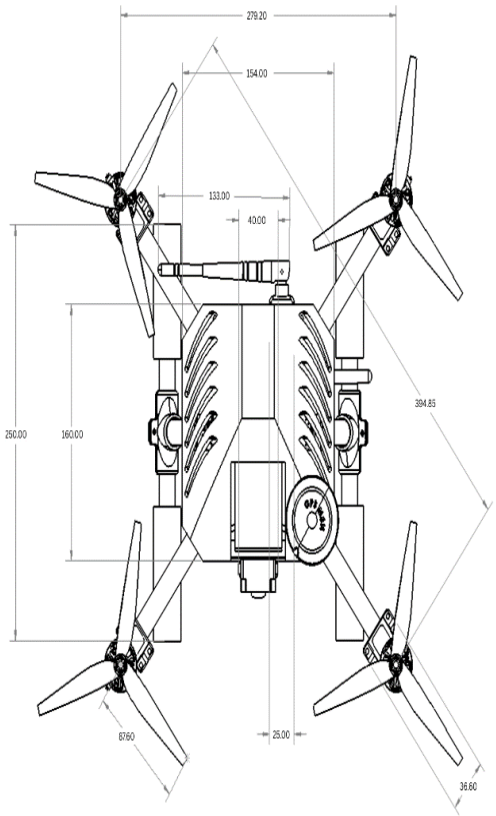
- Precise Payload Deployment
- Immunity to Battery Voltage Fluctuation
- Lightweight Closed-Loop System
- Contactless and Reliable Operation
- Resistant to Dust and Environmental Conditions
- Anti-Jamming Safety Mechanism
- Reduced Power Consumption



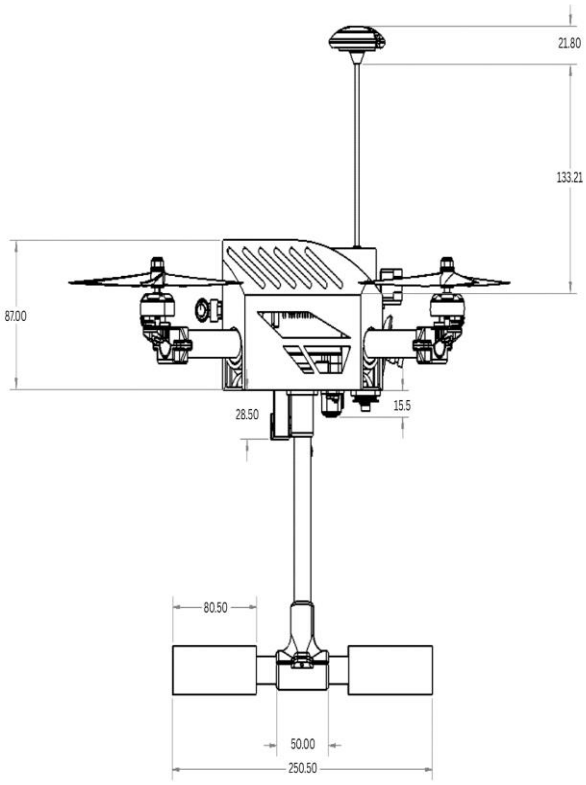




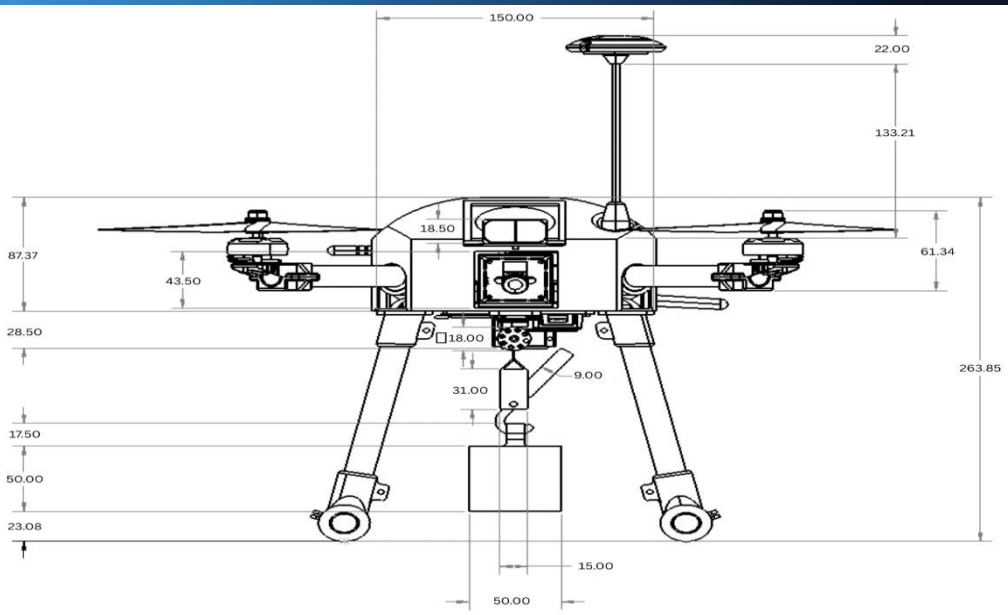
Final UAV design:



Top view



Side view



Front view

Annexures

Annexures 1 – Research Papers and Journals

1. Servo Actuated Payload Carry and Drop Mechanism for Unmanned Helicopter Muhammad Salman Sadiq↤, Umair Iqbal† and Syed Irtiza Ali Shah† ↤Department of Design and Manufacturing School of Mechanical and Manufacturing Engineering National University of Sciences and Technology, Islamabad, Pakistan.

2. Design, Simulation, Analysis and Optimization of PID and Fuzzy Based Control Systems for a Quadcopter Isaac S. Leal 1, Chamil Abeykoon 2, * and Yasith S. Perera 3

3. Mass and 3D centroid test and error analysis of small UAV Liwei Sun1, Shukun Wang2 , Haokun Wang3 , Weihong Zhao4 , Junye Li5 , Liguang Dong6 1, 2, 4, 5, 6Ministry of Education Key Laboratory for Cross-Scale Micro and Nano Manufacturing, Changchun University of Science and Technology, Changchun, China.

4. Adaptive control of a UAV with continuously time varying center of gravity. Shraddha Barawkar ∗ Manish Kumar ∗∗ Michael Bolender ∗∗∗ ∗ University of Cincinnati, 2600 Clifton Ave, Cincinn.

5. ArduPilot Development Team. (2026). *Software-in-the-Loop (SITL) Simulation Overview*. ArduPilot Dev Documentation.

6. Holybro. (2025). *Holybro M10 GPS Module Technical Specification and Integration Guide*. Holybro Docs.

7. Sadiq, M. S., Iqbal, U., & Shah, S. I. A. (2014). Servo actuated payload carry and drop mechanism for unmanned helicopter. *International Journal of Research in Engineering and Technology*, *3*(11), 342–348.

Annexures 2 – Documentations and Community resources.

1. www.researchgate.net
2. in.mathwork.com
3. https://ardupilot.org/dev/docs/sitl-simulator-software-in-the-loop.html
4. https://docs.holybro.com/